What Drives Mortality in Heart Failure? A Fully Bayesian Analysis

Francesco Pio Nardiello, 2251083

2026-06-11

Contents

1 Introduction **3**

1.1 Research Context and Objectives . . . 3

1.2 The Bayesian Framework and Notation . . . 4

2 Illustration of the Dataset **4**

2.1 Data Loading and Overview . . . 4

2.2 Univariate Explorations . . . 7

2.3 Categorical Variable Summaries . . . 9

2.4 Bivariate Relationships . . . 10

2.4.1 Biomarkers vs. Survival . . . 10

2.4.2 Death Rate by Clinical Group . . . 11

2.4.3 Predictor–Outcome Trends . . . 12

2.5 Correlation Matrix . . . 13

2.6 Response and Predictor Preparation . . . 15

3 Models and Inferential Goals **16**

3.1 The Two Competing Models . . . 16

3.2 Prior Specification . . . 16

3.3 JAGS . . . 16

3.4 Role of the Parameters and Inferential Goals . . . 17

3.5 Model Validation: Parameter Recovery on Simulated Data . . . 17

3.5.1 Recovery for the Binomial-Beta model . . . 17

3.5.2 Recovery for the logistic model . . . 18

4 Model 1: Binomial-Beta Conjugate Model **20**

4.1 Model Formulation . . . 20

4.2 Interpretation of the Model . . . 20

4.2.1 Prior vs Posterior . . . 22

4.3 Diagnostic and Posterior Density Plots . . . 23

5 Model 2: Reduced Bayesian Logistic Regression **30**

5.1 Model Formulation . . . 30

5.2 Interpretation of the Model . . . 31

5.2.1 Prior vs Posterior . . . 31

5.3 Diagnostic and Posterior Density Plots . . . 33

6 Model Comparison **40**

7 Posterior Predictive Checks **41**

7.1 Binomial-Beta: Replicated Group Death Counts . . . 41

7.2 Logistic: Replicated Mortality Rate . . . 42

7.3 Logistic: Classification . . . 43

8 Inference and Prediction **44**

8.1 Equal-tailed vs HPD Intervals . . . 44

8.2 Prediction for New Patients and a New Cohort . . . 44

9 Prior Sensitivity Analysis **45**

10 Comparative Analysis: Frequentist vs. Bayesian **46**

10.1 Frequentist Counterparts . . . 46

10.2 Concordance of Findings . . . 48

10.3 Methodological Comparison . . . 48

10.4 Final Comparison . . . 48

11 Conclusions and Key Findings **48**

1 Introduction

This project employs a fully Bayesian framework to study mortality in patients with heart failure. The binary outcome `DEATH_EVENT` (death during follow-up) is the response, and the analysis compares two alternative Bayesian models for the probability of death, both estimated by Markov-chain Monte Carlo. The primary research question is:

> *What drives the probability of death in heart-failure patients, and does a simple model that stratifies the death probability by cardiac function explain the data as well as a richer regression on the main clinical biomarkers?*

The analysis uses data on 299 patients described by 13 variables spanning demographics, comorbidities and laboratory measurements. We deliberately choose **two models that contrast in philosophy** yet share the same Bernoulli response, so that they are directly comparable through the Deviance Information Criterion (DIC):

1. a **Binomial-Beta conjugate model**, the canonical Bayesian model for binary data, which estimates a separate death probability for two clinical groups;
2. a **reduced Bayesian logistic regression**, which models the log-odds of death as a linear function of the four dominant biomarkers.

Under the Bayesian paradigm, the posterior distribution of the parameters is

$$\pi(\theta \mid y) \propto L_y(\theta)\, \pi(\theta),$$

where $\pi(\theta)$ is the prior, $L_y(\theta)$ the likelihood, and the posterior updates prior beliefs in light of the observed outcomes.

1.1 Research Context and Objectives

Heart failure is a leading cause of death worldwide, and identifying the markers that predict mortality is central to clinical risk stratification. Patient outcomes emerge from the interplay of cardiac function (ejection fraction), kidney function (serum creatinine), electrolyte balance (serum sodium), age and comorbidities.

To address the research question, the analysis pursues two objectives. **First**, to compare the two Bayesian specifications above by DIC, determining whether a parsimonious two-group rate model suffices or whether a multivariate regression genuinely improves the description of mortality. **Second**, to estimate and interpret the quantities of clinical interest, the difference in death probability between cardiac-function groups (Binomial-Beta), and the odds ratios of the individual biomarkers (logistic), with full posterior uncertainty. The analysis also delivers Bayesian point estimates, equal-tailed and HPD credible intervals, a prior-vs-posterior comparison, convergence diagnostics (Gelman-Rubin, Geweke, trace plots, ESS, MCSE), model validation through parameter recovery on simulated data, posterior predictive checks, a prior-sensitivity analysis, and a comparison with frequentist inference.

1.2 The Bayesian Framework and Notation

For patient $i = 1, \ldots, N$, let $y_i \in \{0, 1\}$ be the death indicator. The **Binomial-Beta** model assumes $y_i \sim \text{Bernoulli}(\theta_{g_i})$ with $\theta_g \sim \text{Beta}(a, b)$, where g_i is a clinical group; conjugacy gives a Beta posterior in closed form.

The **logistic** model assumes $y_i \sim \text{Bernoulli}(p_i)$ with $\text{logit}(p_i) = \beta_0 + \sum_j \beta_j x_{ij}$; here e^{β_j} is the odds ratio of predictor j.

Both posteriors are explored by MCMC in JAGS, from which every inferential quantity is derived.

2 Illustration of the Dataset

The data are the real clinical records of 299 patients with heart failure, collected at the Faisalabad Institute of Cardiology and the Allied Hospital (Punjab, Pakistan) between April and December 2015. All patients had left-ventricular systolic dysfunction and belonged to NYHA classes III or IV (advanced heart failure). Each patient is described by eleven baseline clinical features plus `time` (follow-up length) and `DEATH_EVENT`. Of the 299 patients, 96 (32%) died and 203 survived.

The variables are:

- **DEATH_EVENT**: binary indicator, 1 if the patient died during follow-up, 0 otherwise (the response).
- **time**: length of the follow-up, in days.
- **age**: patient's age, in years.
- **sex**: binary, 1 for men and 0 for women.
- **smoking**: binary, 1 if the patient smokes.
- **anaemia**: binary, reduced red blood cells or haemoglobin.
- **diabetes**: binary, whether the patient is diabetic.
- **high_blood_pressure**: binary, presence of hypertension.
- **ejection_fraction**: percentage of blood the left ventricle ejects per contraction; the central indicator of severity.
- **serum_creatinine**: serum creatinine (mg/dL); higher values signal worse renal function.
- **platelets**: platelets in the blood (kiloplatelets/mL).
- **serum_sodium**: sodium concentration in the blood (mEq/L).
- **creatinine_phosphokinase**: CPK enzyme level (mcg/L), released when muscle tissue is damaged.

2.1 Data Loading and Overview

```
#setwd("path/to/your/folder")
data <- read.csv("heart_failure_clinical_records_dataset.csv")
cat("Dataset dimensions:", dim(data)[1], "observations x", dim(data)[2], "variables\n\n")
```

```
## Dataset dimensions: 299 observations x 13 variables
```

```r
preview_data <- t(head(data))
colnames(preview_data) <- paste("Obs", 1:6)

knitr::kable(preview_data, caption = "Preview of the first 6 rows")
```

Table 1: Preview of the first 6 rows

	Obs 1	Obs 2	Obs 3	Obs 4	Obs 5	Obs 6
age	75.0	55.0	65.0	50.0	65.0	90.0
anaemia	0.0	0.0	0.0	1.0	1.0	1.0
creatinine_phosphokinase	582.0	7861.0	146.0	111.0	160.0	47.0
diabetes	0.0	0.0	0.0	0.0	1.0	0.0
ejection_fraction	20.0	38.0	20.0	20.0	20.0	40.0
high_blood_pressure	1.0	0.0	0.0	0.0	0.0	1.0
platelets	265000.0	263358.0	162000.0	210000.0	327000.0	204000.0
serum_creatinine	1.9	1.1	1.3	1.9	2.7	2.1
serum_sodium	130.0	136.0	129.0	137.0	116.0	132.0
sex	1.0	1.0	1.0	1.0	0.0	1.0
smoking	0.0	0.0	1.0	0.0	0.0	1.0
time	4.0	6.0	7.0	7.0	8.0	8.0
DEATH_EVENT	1.0	1.0	1.0	1.0	1.0	1.0

```r
str(data)
```

```
## 'data.frame':    299 obs. of  13 variables:
##  $ age                     : num  75 55 65 50 65 90 75 60 65 80 ...
##  $ anaemia                 : int  0 0 0 1 1 1 1 1 0 1 ...
##  $ creatinine_phosphokinase: int  582 7861 146 111 160 47 246 315 157 123 ...
##  $ diabetes                : int  0 0 0 0 1 0 0 1 0 0 ...
##  $ ejection_fraction       : int  20 38 20 20 20 40 15 60 65 35 ...
##  $ high_blood_pressure     : int  1 0 0 0 0 1 0 0 0 1 ...
##  $ platelets               : num  265000 263358 162000 210000 327000 ...
##  $ serum_creatinine        : num  1.9 1.1 1.3 1.9 2.7 2.1 1.2 1.1 1.5 9.4 ...
##  $ serum_sodium            : int  130 136 129 137 116 132 137 131 138 133 ...
##  $ sex                     : int  1 1 1 1 0 1 1 1 0 1 ...
##  $ smoking                 : int  0 0 1 0 0 1 0 1 0 1 ...
##  $ time                    : int  4 6 7 7 8 8 10 10 10 10 ...
##  $ DEATH_EVENT             : int  1 1 1 1 1 1 1 1 1 1 ...
```

```r
summary(data)
```

```
##       age           anaemia       creatinine_phosphokinase    diabetes
```

```
##  Min.   :40.00   Min.   :0.0000   Min.   :  23.0          Min.   :0.0000
##  1st Qu.:51.00   1st Qu.:0.0000   1st Qu.: 116.5          1st Qu.:0.0000
##  Median :60.00   Median :0.0000   Median : 250.0          Median :0.0000
##  Mean   :60.83   Mean   :0.4314   Mean   : 581.8          Mean   :0.4181
##  3rd Qu.:70.00   3rd Qu.:1.0000   3rd Qu.: 582.0          3rd Qu.:1.0000
##  Max.   :95.00   Max.   :1.0000   Max.   :7861.0          Max.   :1.0000
##  ejection_fraction high_blood_pressure   platelets      serum_creatinine
##  Min.   :14.00     Min.   :0.0000      Min.   : 25100   Min.   :0.500
##  1st Qu.:30.00     1st Qu.:0.0000      1st Qu.:212500   1st Qu.:0.900
##  Median :38.00     Median :0.0000      Median :262000   Median :1.100
##  Mean   :38.08     Mean   :0.3512      Mean   :263358   Mean   :1.394
##  3rd Qu.:45.00     3rd Qu.:1.0000      3rd Qu.:303500   3rd Qu.:1.400
##  Max.   :80.00     Max.   :1.0000      Max.   :850000   Max.   :9.400
##   serum_sodium        sex            smoking            time
##  Min.   :113.0   Min.   :0.0000   Min.   :0.0000   Min.   :  4.0
##  1st Qu.:134.0   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.: 73.0
##  Median :137.0   Median :1.0000   Median :0.0000   Median :115.0
##  Mean   :136.6   Mean   :0.6488   Mean   :0.3211   Mean   :130.3
##  3rd Qu.:140.0   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:203.0
##  Max.   :148.0   Max.   :1.0000   Max.   :1.0000   Max.   :285.0
##   DEATH_EVENT
##  Min.   :0.0000
##  1st Qu.:0.0000
##  Median :0.0000
##  Mean   :0.3211
##  3rd Qu.:1.0000
##  Max.   :1.0000
```

```r
sum(is.na(data))
```

```
## [1] 0
```

Interpretation. The cohort is composed of seriously ill patients: the average ejection fraction of about 38%, well below the healthy 55–70% range, confirms substantially impaired cardiac function. Ages span 40–95 and follow-up averages roughly four months. About one in three (96/299, 32%) died. The dataset is complete (there are not missing values), so no imputation is needed, and the 32%/68% split is well balanced for modelling.

```r
data$outcome <- factor(data$DEATH_EVENT, levels = c(0, 1), labels = c("Survived", "Died"))
cont_vars <- c("age", "creatinine_phosphokinase", "ejection_fraction",
               "platelets", "serum_creatinine", "serum_sodium", "time")
skew <- function(x) mean((x - mean(x))^3) / mean((x - mean(x))^2)^1.5
```

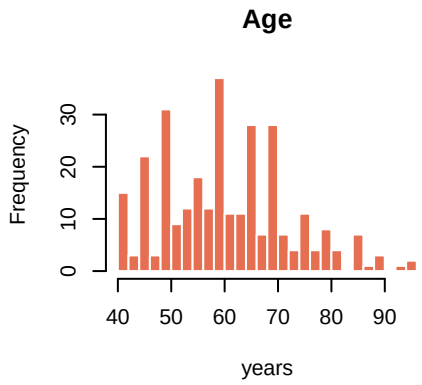
2.2 Univariate Explorations

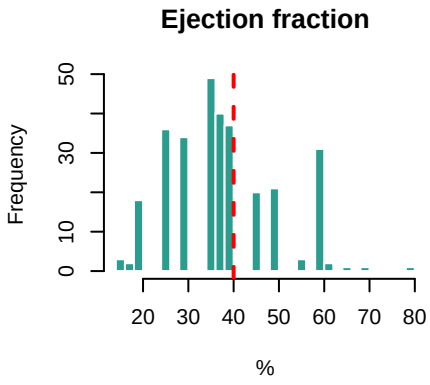
We examine the marginal distributions of the numeric variables to identify skewness, outliers and central tendencies.

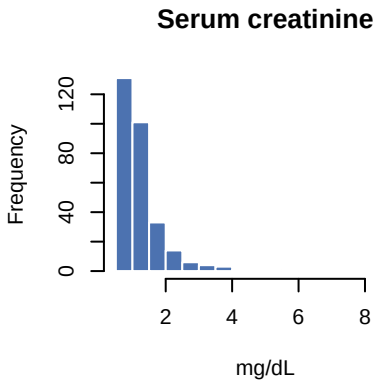
```
round(t(sapply(cont_vars, function(v)
  c(min = min(data[[v]]), median = median(data[[v]]), mean = mean(data[[v]]),
    max = max(data[[v]]), sd = sd(data[[v]]), skew = skew(data[[v]])))), 2)
```

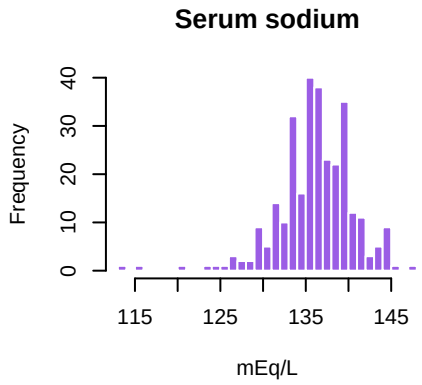
```
##                              min   median      mean      max       sd  skew
## age                         40.0     60.0     60.83     95.0    11.89  0.42
## creatinine_phosphokinase    23.0    250.0    581.84   7861.0   970.29  4.44
## ejection_fraction           14.0     38.0     38.08     80.0    11.83  0.55
## platelets                25100.0 262000.0 263358.03 850000.0 97804.24  1.45
## serum_creatinine             0.5      1.1      1.39      9.4     1.03  4.43
## serum_sodium               113.0    137.0    136.63    148.0     4.41 -1.04
## time                         4.0    115.0    130.26    285.0    77.61  0.13
```

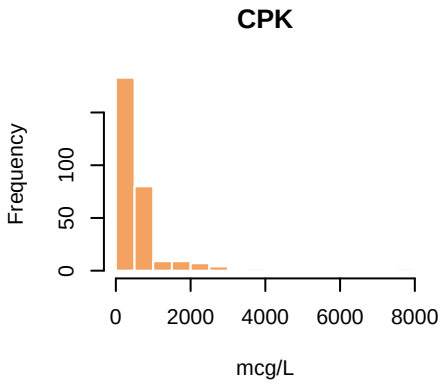
```
par(mfrow = c(2, 3))
hist(data$age, breaks = 25, col = "#e76f51", border = "white", main = "Age", xlab = "years")
hist(data$ejection_fraction, breaks = 25, col = "#2a9d8f", border = "white",
     main = "Ejection fraction", xlab = "%"); abline(v = 40, col = "red", lwd = 2, lty = 2)
hist(data$serum_creatinine, breaks = 25, col = "#4c72b0", border = "white",
     main = "Serum creatinine", xlab = "mg/dL")
hist(data$serum_sodium, breaks = 25, col = "#9b5de5", border = "white",
     main = "Serum sodium", xlab = "mEq/L")
hist(data$creatinine_phosphokinase, breaks = 25, col = "#f4a261", border = "white",
     main = "CPK", xlab = "mcg/L")
hist(data$platelets, breaks = 25, col = "#90be6d", border = "white",
     main = "Platelets", xlab = "kiloplatelets/mL")
```

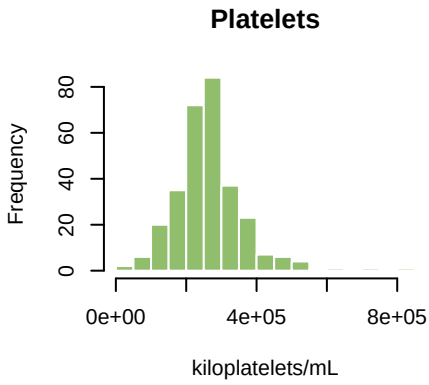








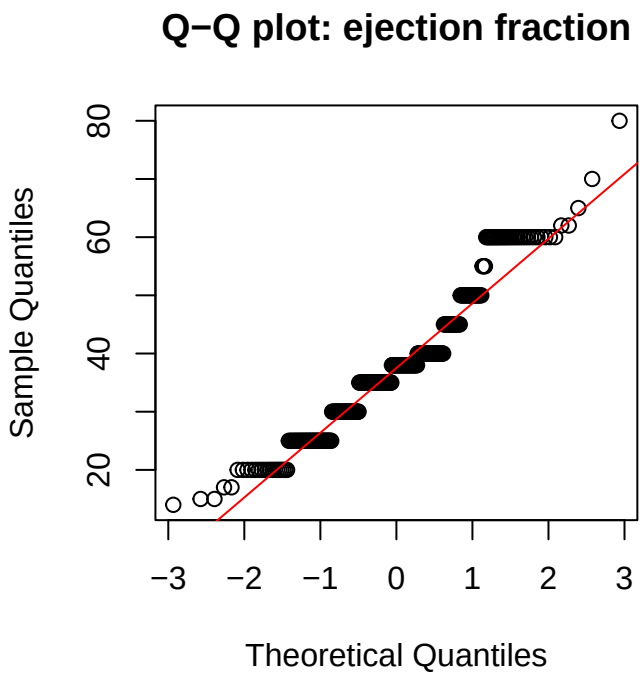


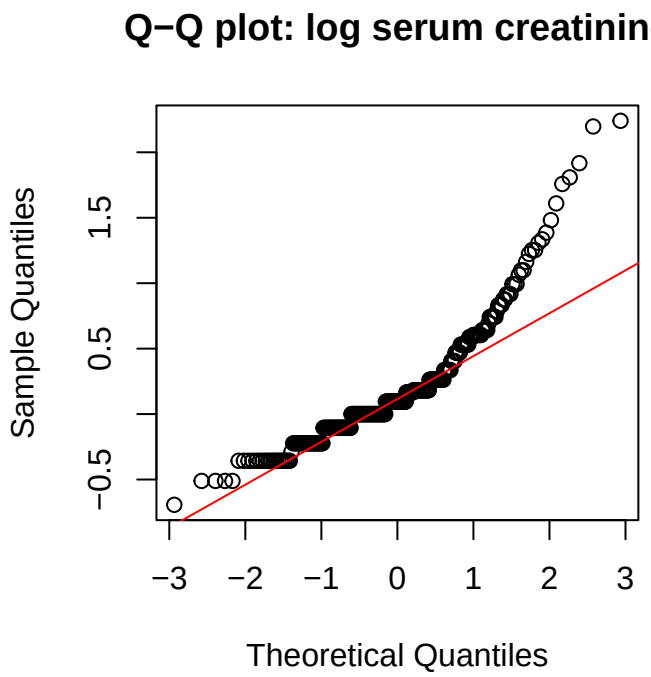



```
par(mfrow = c(1, 1))
```

We also assess the normality of the central biomarker with a Q-Q plot.

```
par(mfrow = c(1, 2))
qqnorm(data$ejection_fraction, main = "Q-Q plot: ejection fraction"); qqline(data$ejection_fraction, col
qqnorm(log(data$serum_creatinine), main = "Q-Q plot: log serum creatinine"); qqline(log(data$serum_creat
```






```
par(mfrow = c(1, 1))
```

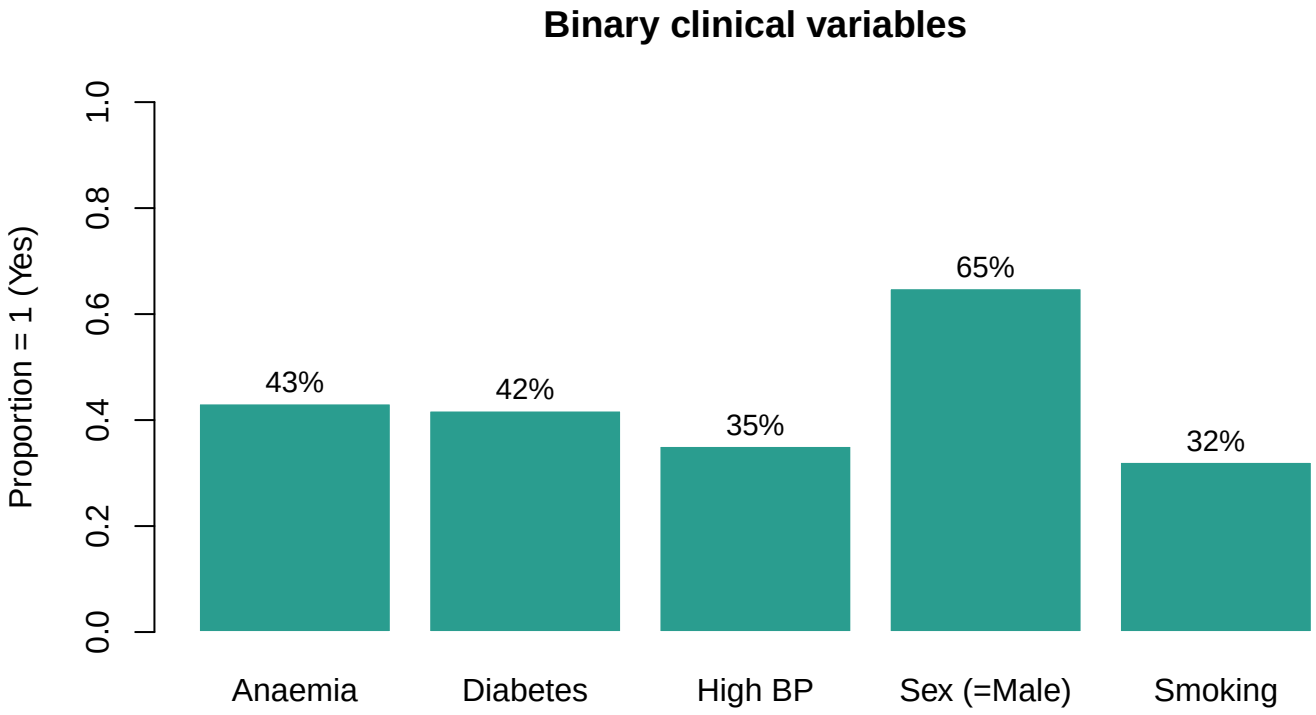
Key observations. Age is roughly symmetric around 60; ejection fraction is mildly right-skewed around 38%; serum sodium is symmetric and bell-shaped; serum creatinine, CPK and platelets are strongly right-skewed. The Q-Q plots show ejection fraction is approximately normal while serum creatinine is far closer to normality on the log scale – a fact we exploit by **log-transforming serum creatinine before standardising it** for the regression (Section 2.6).

2.3 Categorical Variable Summaries

```
bin_pred <- c("anaemia", "diabetes", "high_blood_pressure", "sex", "smoking")
props <- sapply(bin_pred, function(v) mean(data[[v]])); round(props, 3)
```

```
##             anaemia            diabetes high_blood_pressure                 sex
##               0.431               0.418               0.351               0.649
##             smoking
##               0.321
```

```
bp <- barplot(props, ylim = c(0, 1), col = "#2a9d8f", border = "white",
              names.arg = c("Anaemia", "Diabetes", "High BP", "Sex (=Male)", "Smoking"),
              ylab = "Proportion = 1 (Yes)", main = "Binary clinical variables")
text(bp, props + 0.04, labels = sprintf("%.0f%%", 100 * props), cex = 0.9)
```



Key observations. The binary comorbidities are reasonably balanced (anaemia 43%, diabetes 42%, high blood pressure 35%, smoking 32%; 65% male), so none dominates the cohort and each carries estimable variation.

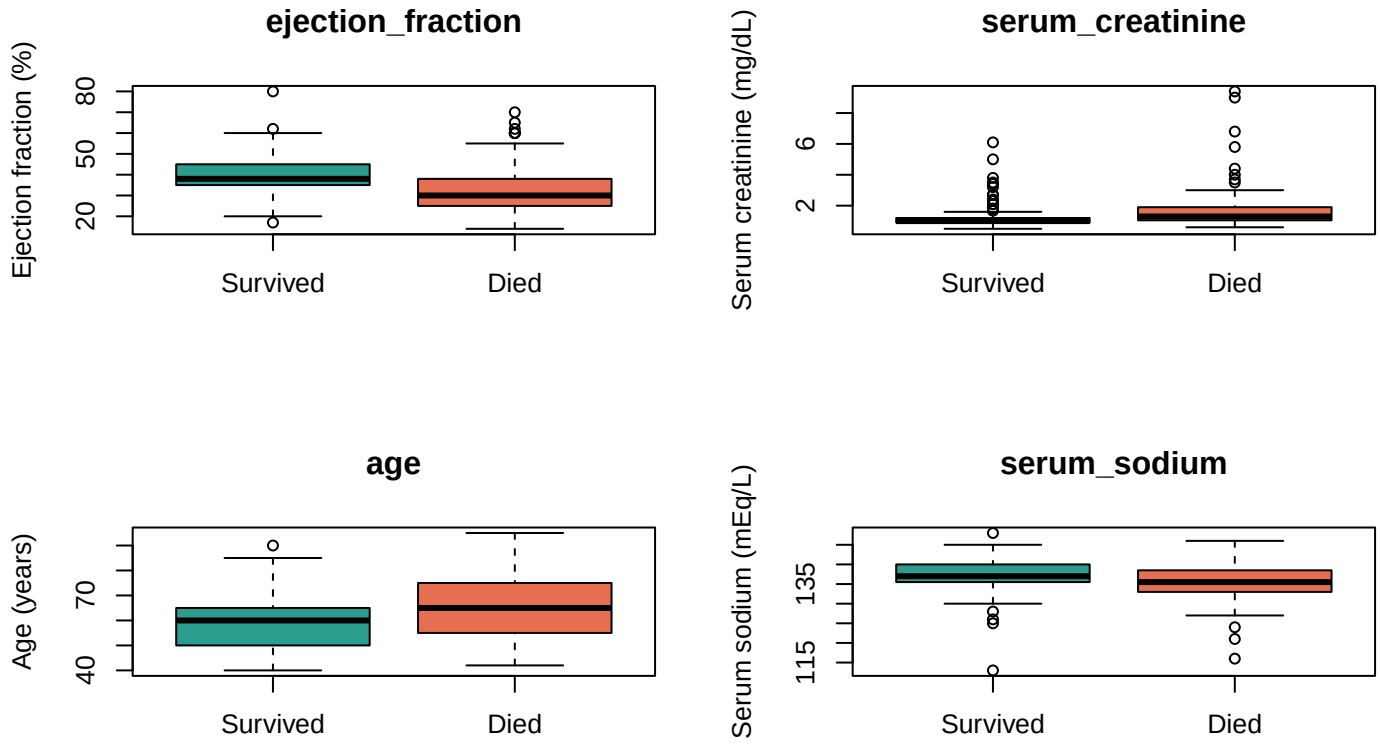
2.4 Bivariate Relationships

2.4.1 Biomarkers vs. Survival

```
key  <- c("ejection_fraction", "serum_creatinine", "age", "serum_sodium")
labs <- c("Ejection fraction (%)", "Serum creatinine (mg/dL)", "Age (years)", "Serum sodium (mEq/L)")
sapply(key, function(v) round(tapply(data[[v]], data$outcome, mean), 2))
```

```
##          ejection_fraction serum_creatinine   age serum_sodium
## Survived             40.27             1.18 58.76       137.22
## Died                 33.47             1.84 65.22       135.38
```

```r
par(mfrow = c(2, 2))
for (i in seq_along(key))
  boxplot(data[[key[i]]] ~ data$outcome, col = c("#2a9d8f", "#e76f51"),
          main = key[i], ylab = labs[i], xlab = "")
```




```r
par(mfrow = c(1, 1))
```

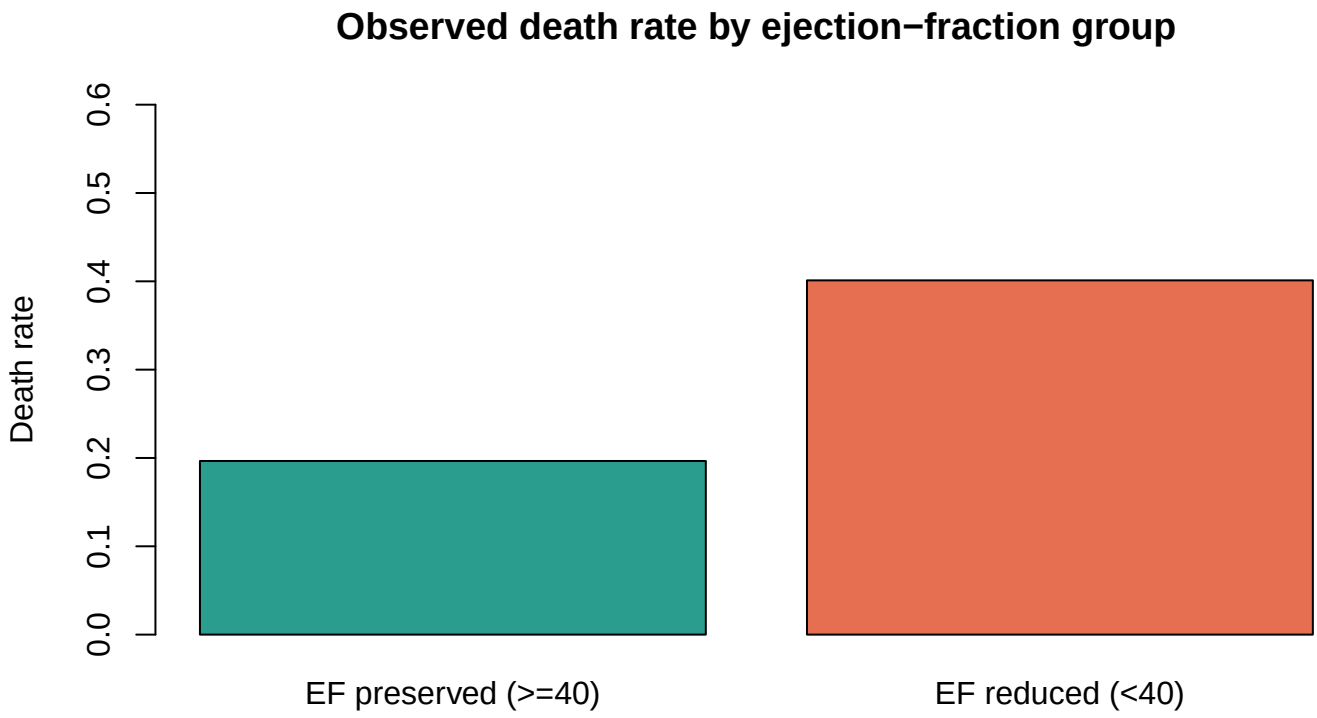
Interpretation. Patients who died had lower ejection fraction (33.5 vs 40.3), higher serum creatinine (1.84 vs 1.18), older age (65.2 vs 58.8) and slightly lower serum sodium (135.4 vs 137.2), all in the clinically expected direction.

2.4.2 Death Rate by Clinical Group

```r
data$grp_ef_lab <- ifelse(data$ejection_fraction < 40, "EF reduced (<40)", "EF preserved (>=40)")
round(tapply(data$DEATH_EVENT, data$grp_ef_lab, mean), 3)
```

```
## EF preserved (>=40)    EF reduced (<40)
##               0.197               0.401
```

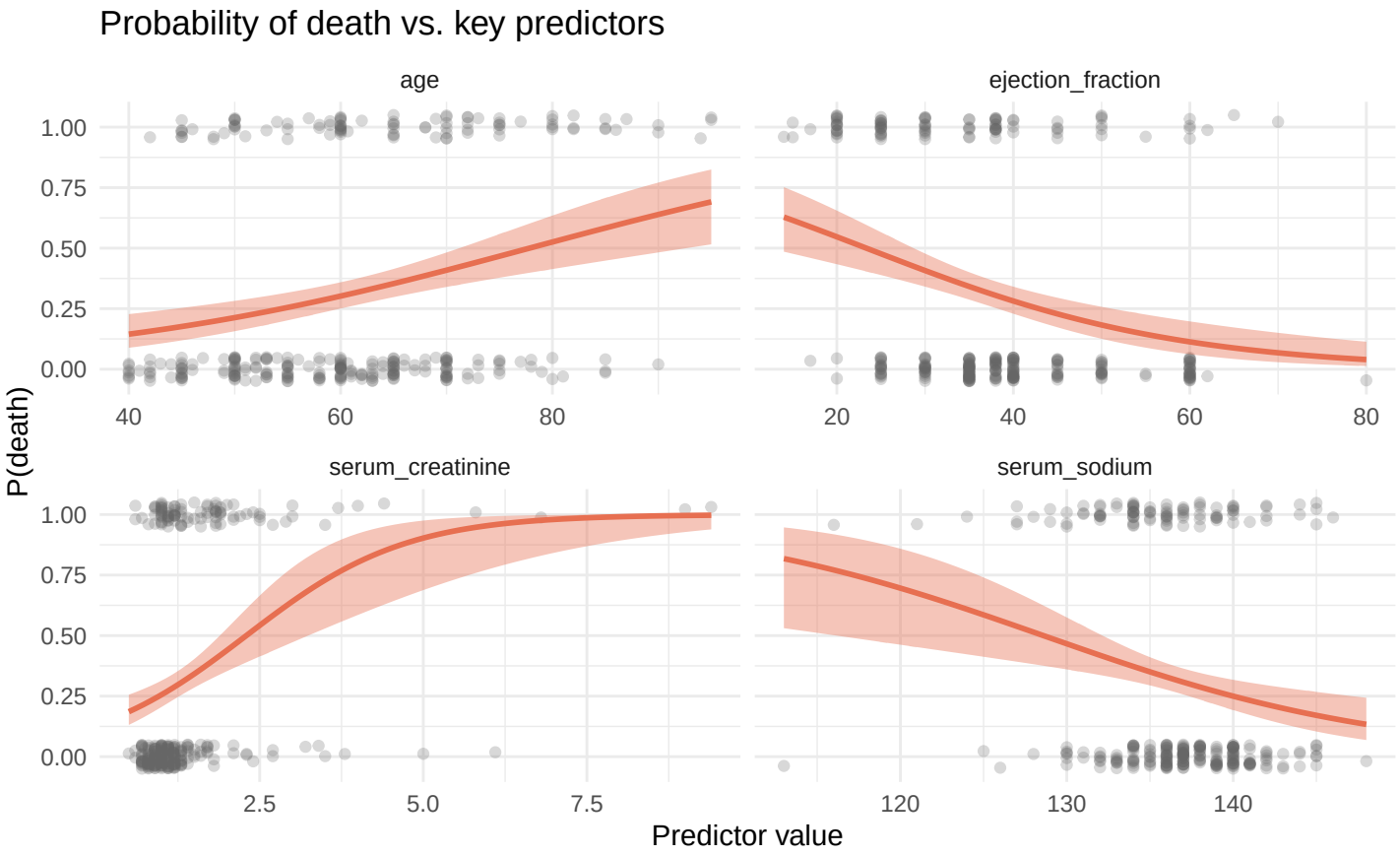
```r
barplot(tapply(data$DEATH_EVENT, data$grp_ef_lab, mean), col = c("#2a9d8f", "#e76f51"),
        ylim = c(0, 0.6), ylab = "Death rate", main = "Observed death rate by ejection-fraction group")
```



Interpretation. The observed death rate is markedly higher in the reduced-ejection-fraction group than in the preserved-EF group, precisely the contrast the Binomial-Beta model will formalise.

2.4.3 Predictor–Outcome Trends

```r
df_long <- data %>% select(DEATH_EVENT, all_of(key)) %>%
  pivot_longer(all_of(key), names_to = "predictor", values_to = "value")
ggplot(df_long, aes(value, DEATH_EVENT)) +
  geom_jitter(height = 0.05, width = 0, alpha = 0.25, colour = "grey40") +
  geom_smooth(method = "glm", method.args = list(family = "binomial"),
              se = TRUE, colour = "#e76f51", fill = "#e76f51") +
  facet_wrap(~ predictor, scales = "free_x") +
  labs(title = "Probability of death vs. key predictors", x = "Predictor value", y = "P(death)") +
  theme_minimal()
```



Interpretation. The fitted logistic smoothers rise with age and serum creatinine and fall with ejection fraction and serum sodium, a visual preview of the regression coefficient signs.

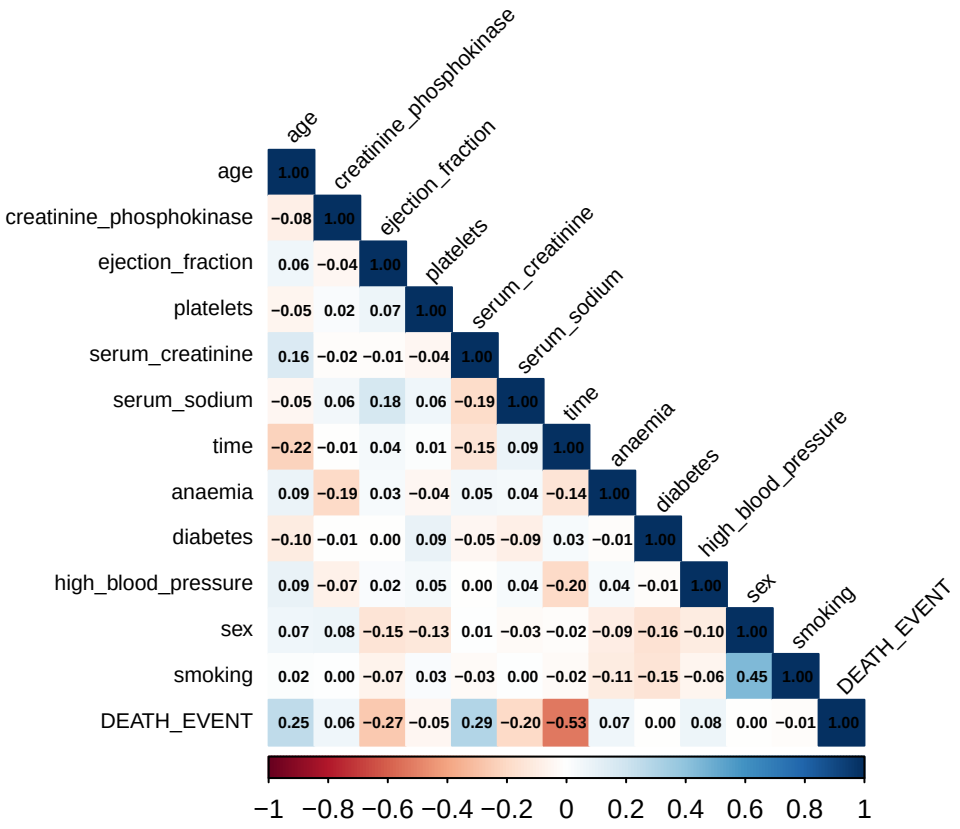
2.5 Correlation Matrix

```
num <- data[, c(cont_vars, "anaemia", "diabetes", "high_blood_pressure", "sex", "smoking", "DEATH_EVENT
C <- cor(num)
round(sort(C[, "DEATH_EVENT"], decreasing = TRUE), 3)
```

```
##              DEATH_EVENT         serum_creatinine                      age
##                    1.000                    0.294                    0.254
##      high_blood_pressure                  anaemia creatinine_phosphokinase
##                    0.079                    0.066                    0.063
##                 diabetes                      sex                  smoking
##                   -0.002                   -0.004                   -0.013
##                platelets             serum_sodium        ejection_fraction
##                   -0.049                   -0.195                   -0.269
##                     time
```

```
##                   -0.527
```

```
corrplot(C, method = "color", type = "lower", order = "original",
         addCoef.col = "black", number.cex = 0.5, tl.col = "black", tl.srt = 45, tl.cex = 0.7)
```



Insights. The correlation matrix displays that the strongest association is with time (-0.53): patients with shorter follow-up are far more likely to have died. This correlation, however, is an artefact of the study design rather than a risk factor becuase follow-up necessarily ends at death, which is why time is excluded from the predictors. Setting it aside, the variables most strongly correlated with mortality are serum creatinine (+0.29), ejection fraction (-0.27), age (+0.25) and serum sodium (-0.20). Their signs are clinically coherent: worse kidney function and older age raise the probability of death, whereas a higher ejection fraction (a stronger heart) and higher serum sodium lower it. All remaining covariates like creatinine phosphokinase, platelets, anaemia, diabetes, high blood pressure, sex and smoking, have correlations near zero with the outcome (between -0.05 and +0.08), indicating essentially no marginal association.

Among the predictors themselves, the correlations are uniformly weak, which is reassuring as it signals little multicollinearity. The only notable exception is the moderate positive correlation between smoking and sex (+0.45), reflecting the fact that smoking is far more prevalent among the men in this cohort. A few other mild, clinically sensible relationships appea, for instance serum sodium correlates slightly positively with

ejection fraction (+0.18) and negatively with serum creatinine (-0.19), but none is strong enough to cause estimation problems.

Overall, the matrix anticipates the modelling results: a small set of physiological markers carries the predictive signal for mortality, while the demographic and comorbidity variables contribute little, motivating the move from the full model to the reduced one.

2.6 Response and Predictor Preparation

The response `DEATH_EVENT` is binary and enters both Bernoulli likelihoods unchanged. For the **logistic** model we prepare the four continuous predictors so that coefficients are comparable (in log-odds per standard deviation) and the sampler is stable. Because serum creatinine is strongly right-skewed (skewness ≈ 4.4), we first **log-transform** it – which brings it close to normality, as the Q-Q plot showed – and then standardise it; the other three predictors are standardised on their natural scale. All four end up with mean 0 and standard deviation 1. For the **Binomial-Beta** model we form a clinical group from the ejection-fraction cutoff (EF $< 40\%$). The variable `time` is excluded from the predictors: although it correlates strongly with death, it is measured after the outcome unfolds (a patient who dies early necessarily has short follow-up), so including it would leak the outcome into the model. We therefore use the eleven baseline clinical covariates only.

```
yv <- data$DEATH_EVENT
N  <- length(yv)

# logistic design (4 predictors). serum_creatinine is highly right-skewed
# (skew ~ 4.4), so we LOG-transform it BEFORE standardising; this brings it
# close to normality. The other three are standardised on
# their natural scale. All four predictors end up with mean 0 and sd 1.
data_std <- data
data_std$serum_creatinine <- log(data$serum_creatinine)
std_cols <- c("age", "ejection_fraction", "serum_creatinine", "serum_sodium")
data_std[std_cols] <- scale(data_std[std_cols])
f_red <- DEATH_EVENT ~ age + ejection_fraction + serum_creatinine + serum_sodium
X_red <- model.matrix(f_red, data_std)[, -1]
round(colMeans(X_red), 6); round(apply(X_red, 2, sd), 6)
```

```
##               age ejection_fraction  serum_creatinine      serum_sodium
##                 0                 0                 0                 0

##               age ejection_fraction  serum_creatinine      serum_sodium
##                 1                 1                 1                 1
```

```
# Binomial-Beta grouping (1 = reduced EF, 2 = preserved)
grp_ef <- ifelse(data$ejection_fraction < 40, 1, 2)
table(grp_ef)
```

```
## grp_ef
##   1   2
## 182 117
```

3 Models and Inferential Goals

3.1 The Two Competing Models

Model 1: Binomial-Beta conjugate model. Patients are split by reduced ejection fraction, and each group has its own death probability,

$$y_i \sim \text{Bernoulli}(\theta_{g_i}), \qquad \theta_1, \theta_2 \sim \text{Beta}(1, 1),$$

with θ_1 for reduced EF, θ_2 for preserved EF, and effect of interest $\Delta = \theta_1 - \theta_2$. This is the conjugate model for binary data; the posterior is a Beta in closed form.

Model 2: Reduced Bayesian logistic regression. The log-odds of death depend linearly on the four dominant biomarkers,

$$\text{logit}(p_i) = \beta_0 + \beta_{\text{age}} x_{\text{age}} + \beta_{\text{ef}} x_{\text{ef}} + \beta_{\text{screat}} x_{\text{screat}} + \beta_{\text{sodium}} x_{\text{sodium}}.$$

Both model the same Bernoulli response, so their DIC is directly comparable: the comparison asks whether the continuous regression buys predictive accuracy over the simple two-group rate.

3.2 Prior Specification

We use weakly informative priors so the data dominate. For Model 1, $\theta_g \sim \text{Beta}(1, 1)$, the uniform prior on $(0, 1)$. For Model 2, $\beta_0, \beta_j \sim \mathcal{N}(0, 10^3)$ (precision 10^{-3}): on the standardised scale a coefficient of ± 3 already corresponds to an odds ratio of ≈ 20, so this prior is essentially flat over the plausible range. Both choices are stress-tested in the prior-sensitivity analysis.

3.3 JAGS

Both models are written as ASCII files and estimated in JAGS with **3 chains of 20,000 iterations and a 5,000 burn-in**, monitored by $\hat{R}$, ESS, trace plots and the Geweke diagnostic.

```
cat("
model{
  for(i in 1:N){ y[i] ~ dbern(theta[grp[i]]) }
  for(k in 1:G){ theta[k] ~ dbeta(1, 1) }
}
", file = "model_betabin.txt")
```

```
cat("
model{
  for(i in 1:N){
    y[i] ~ dbern(p[i])
    logit(p[i]) <- b0 + inprod(X[i,], beta[])
  }
  b0 ~ dnorm(0, 0.001)
  for(j in 1:K){ beta[j] ~ dnorm(0, 0.001) }
}
", file = "model_logit.txt")
```

3.4 Role of the Parameters and Inferential Goals

In Model 1, θ_1 and θ_2 are the death probabilities of the two cardiac-function groups, and Δ measures the absolute increase in death probability due to reduced ejection fraction. In Model 2, β_0 is the baseline log-odds for an average patient and each β_j the adjusted effect of a biomarker, with e^{β_j} its odds ratio. The inferential goals are to estimate these quantities with posterior means and HPD intervals, to assess the hypotheses $P(\Delta > 0 \mid y)$ and $P(\beta_j > 0 \mid y)$, to predict the outcome for new patients/cohorts, to compare the two models by DIC, and to validate them through parameter recovery and posterior predictive checks.

3.5 Model Validation: Parameter Recovery on Simulated Data

Before the real data, we verify that both models recover known parameters.

3.5.1 Recovery for the Binomial-Beta model

```
set.seed(123)
n1 <- 150; n2 <- 149; th1 <- 0.45; th2 <- 0.18
y_s <- c(rbinom(n1, 1, th1), rbinom(n2, 1, th2)); g_s <- c(rep(1, n1), rep(2, n2))
rec_bb <- jags(list(y = y_s, grp = g_s, N = n1 + n2, G = 2),
               parameters.to.save = "theta", model.file = "model_betabin.txt",
               n.chains = 3, n.iter = 20000, n.burnin = 5000)
```

```
sb <- rec_bb$BUGSoutput$sims.matrix
knitr::kable(data.frame(parameter = c("theta[1]", "theta[2]"), true = c(th1, th2),
                        post_mean = colMeans(sb[, c("theta[1]", "theta[2]")]),
                        HPDinterval(as.mcmc(sb[, c("theta[1]", "theta[2]")]))),
             digits = 3, row.names = FALSE, caption = "Binomial-Beta parameter recovery")
```

Table 2: Binomial-Beta parameter recovery

parameter	true	post_mean	lower	upper
theta[1]	0.45	0.449	0.377	0.533
theta[2]	0.18	0.172	0.111	0.228

Interpretation: We first test the Binomial-Beta machinery on data we generate ourselves. We simulate two groups of comparable size to the real ones (150 and 149 patients) with chosen true death probabilities, $\theta_1 = 0.45$ for the high-risk group and $\theta_2 = 0.18$ for the low-risk group, and fit the same two-group model. Because we know the truth, we can check whether the posterior recovers it. The table reports, for each group, the true value, the posterior mean and the 95% HPD interval: the posterior means land close to the generating values, and both HPD intervals comfortably contain the true probability. Since the difference $\theta_1 - \theta_2$ is also recovered far from zero, the model correctly detects the group gap it was designed to estimate. This confirms that the Binomial-Beta sampler is reliable before we apply it to the real data.

3.5.2 Recovery for the logistic model

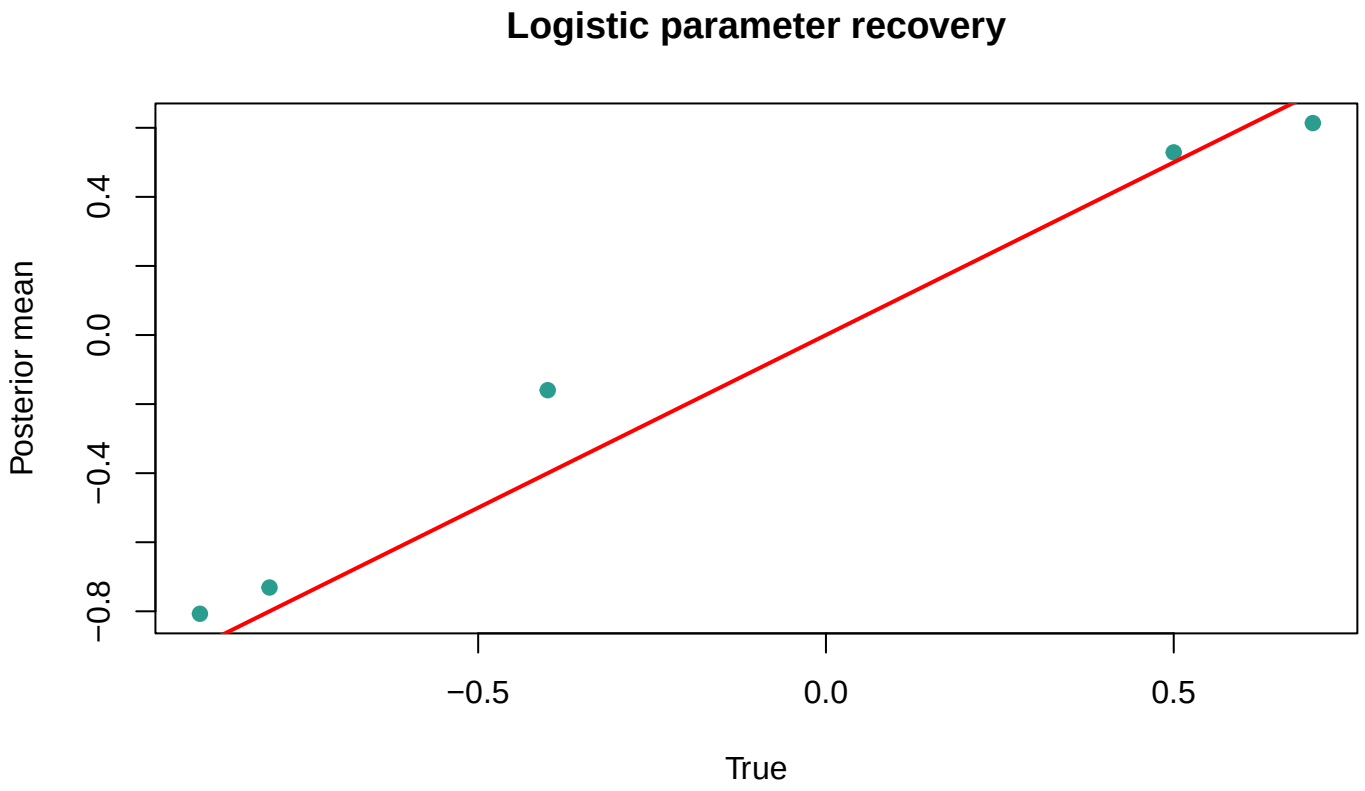
```
Kr <- ncol(X_red)            # kept for the real model fitted later
# A stringent recovery test: instead of reusing the real design matrix, we
# simulate a BRAND-NEW, independent set of covariates from standard Normals,
# so we test the logistic machinery itself rather than the specific sample.
set.seed(321)
Ksim <- 4
Xsim <- matrix(rnorm(N * Ksim), nrow = N, ncol = Ksim)
colnames(Xsim) <- c("x1", "x2", "x3", "x4")
beta_true <- c(x1 = 0.5, x2 = -0.9, x3 = 0.7, x4 = -0.4)
b0_true <- -0.8
p_s  <- inv.logit(as.numeric(b0_true + Xsim %*% beta_true))
y_sl <- rbinom(N, 1, p_s)
rec_lg <- jags(list(y = y_sl, X = Xsim, N = N, K = Ksim),
               parameters.to.save = c("b0", "beta"), model.file = "model_logit.txt",
               n.chains = 3, n.iter = 20000, n.burnin = 5000)
```

```
sl <- rec_lg$BUGSoutput$sims.matrix
recl <- data.frame(parameter = c("b0", colnames(Xsim)),
                   true = c(b0_true, beta_true),
                   post_mean = c(mean(sl[, "b0"]), colMeans(sl[, paste0("beta[", 1:Ksim, "]")])))
hl <- HPDinterval(as.mcmc(sl[, c("b0", paste0("beta[", 1:Ksim, "]"))]))
recl$lwr <- hl[, 1]; recl$upr <- hl[, 2]
knitr::kable(recl, digits = 3, row.names = FALSE, caption = "Logistic parameter recovery")
```

Table 3: Logistic parameter recovery

parameter	true	post_mean	lwr	upr
b0	-0.8	-0.731	-1.016	-0.454
x1	0.5	0.529	0.241	0.811
x2	-0.9	-0.807	-1.113	-0.513
x3	0.7	0.614	0.333	0.887
x4	-0.4	-0.160	-0.420	0.095

```
plot(recl$true, recl$post_mean, pch = 19, col = "#2a9d8f",
     xlab = "True", ylab = "Posterior mean", main = "Logistic parameter recovery"); abline(0, 1, col =
```



Interpretation. We repeat the procedure for the logistic model with a deliberately stringent design: rather than reusing the observed covariates, we simulate a **brand-new, independent set of four predictors** from standard Normal distributions. This tests the Bayesian logistic machinery itself – its ability to recover known coefficients on data generated entirely from the model – rather than merely checking that JAGS runs on our particular sample. We fix a sparse vector of true coefficients ($\beta = (0.5, -0.9, 0.7, -0.4)$, baseline log-odds -0.8), compute the implied death probabilities through the inverse logit, and draw binary outcomes

from them. Fitting the model to these synthetic data, the posterior means align with the true coefficients along the 45-degree line of the diagnostic plot and the credible intervals cover the true values, with the recovered signs and magnitudes matching those imposed. Because the covariates are independent of the real dataset, the successful recovery shows the logistic structure works in general, not just on the observed sample. With both models validated on data of known structure, we can trust the posterior estimates obtained on the real patients in the following sections.

4 Model 1: Binomial-Beta Conjugate Model

4.1 Model Formulation

As above, $y_i \sim \text{Bernoulli}(\theta_{g_i})$ with $\theta_g \sim \text{Beta}(1,1)$, group 1 = reduced EF, group 2 = preserved EF. Conjugacy gives $\theta_g \mid y \sim \text{Beta}(1+d_g, 1+n_g-d_g)$, used below as an exact benchmark.

```
M_bb <- jags(list(y = yv, grp = grp_ef, N = N, G = 2),
             parameters.to.save = "theta", model.file = "model_betabin.txt",
             n.chains = 3, n.iter = 20000, n.burnin = 5000)
```

```
print(M_bb)
```

```
## Inference for Bugs model at "model_betabin.txt", fit using jags,
##  3 chains, each with 20000 iterations (first 5000 discarded), n.thin = 15
##  n.sims = 3000 iterations saved. Running time = 0.17 secs
##           mu.vect sd.vect    2.5%     25%     50%     75%   97.5%  Rhat n.eff
## theta[1]   0.402   0.036   0.332   0.377   0.401   0.427   0.475 1.001  3000
## theta[2]   0.201   0.036   0.135   0.176   0.199   0.224   0.273 1.001  3000
## deviance 363.094   1.913 361.168 361.701 362.506 363.855 368.250 1.002  1300
##
## For each parameter, n.eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor (at convergence, Rhat=1).
##
## DIC info (using the rule: pV = var(deviance)/2)
## pV = 1.8 and DIC = 364.9
## DIC is an estimate of expected predictive error (lower deviance is better).
```

4.2 Interpretation of the Model

Fitting the two-group Binomial-Beta model to the real data gives a posterior mean death probability of about 0.40 in the reduced-ejection-fraction group (θ_1, 95% credible interval roughly 0.33 – 0.48) against about 0.20 in the preserved-EF group (θ_2, interval roughly 0.14 - 0.28). The two intervals barely overlap, a first clear sign that mortality is markedly higher when cardiac function is impaired: patients with reduced ejection fraction die at roughly twice the rate of those with preserved function. The convergence indicators

printed alongside the estimates are excellent, the potential scale reduction factor is $\hat{R} = 1.001$ for both probabilities and the effective sample size reaches the full 3000 saved draws, confirming that the chains have mixed properly and the posterior summaries are reliable.

The model uses an effective number of parameters $p_D = 2.0$, exactly the two group probabilities it estimates, and yields a DIC of 365.0, the reference value against which the logistic model will be compared.

```
s_bb <- M_bb$BUGSoutput$sims.matrix
delta_bb <- s_bb[, "theta[1]"] - s_bb[, "theta[2]"]
tab_bb <- rbind(
  "theta reduced EF" = c(mean = mean(s_bb[, "theta[1]"]), HPDinterval(as.mcmc(s_bb[, "theta[1]"]))),
  "theta preserved"  = c(mean = mean(s_bb[, "theta[2]"]), HPDinterval(as.mcmc(s_bb[, "theta[2]"]))),
  "delta = t1 - t2"  = c(mean = mean(delta_bb), HPDinterval(as.mcmc(delta_bb))))
knitr::kable(round(tab_bb, 3), caption = "Binomial-Beta: posterior summaries (95% HPD)")
```

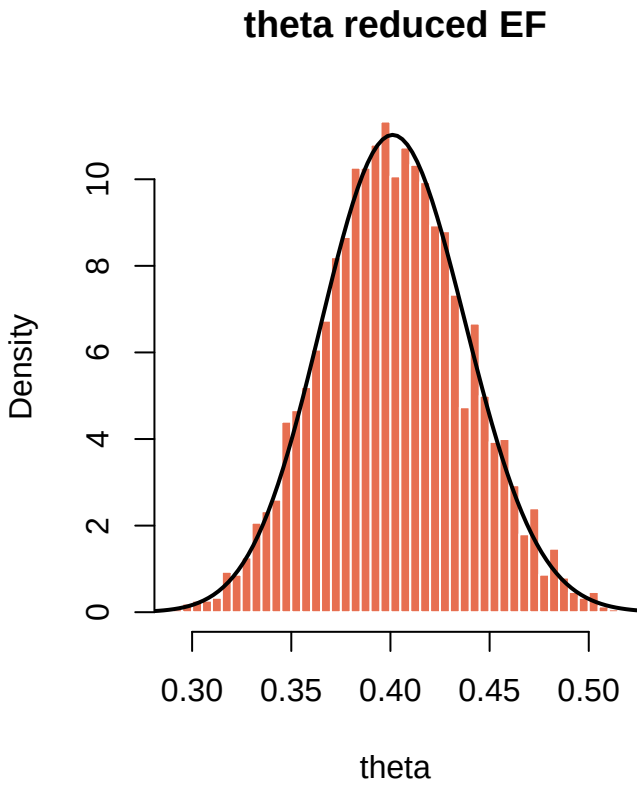
Table 4: Binomial-Beta: posterior summaries (95% HPD)

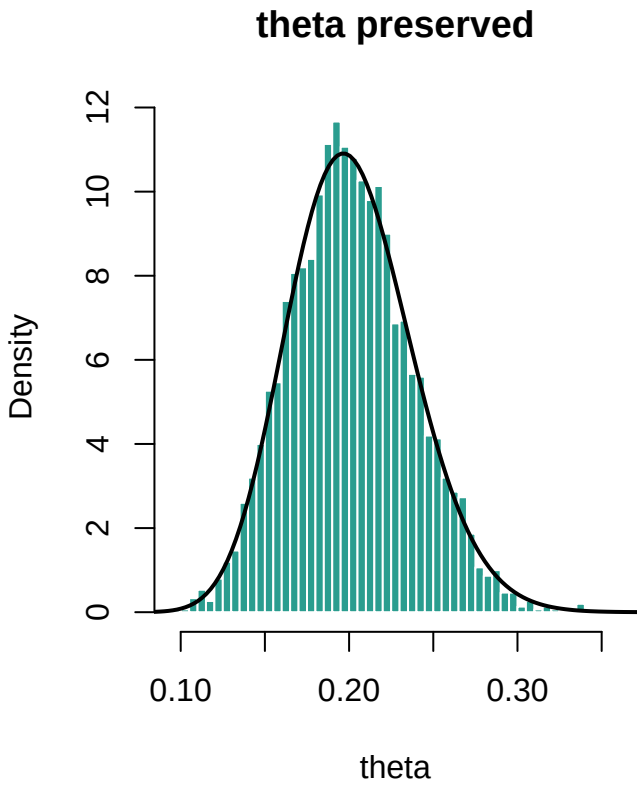
	mean		
theta reduced EF	0.402	0.333	0.475
theta preserved	0.201	0.135	0.273
delta = t1 - t2	0.201	0.101	0.304

```
cat("P(delta > 0 | data) =", round(mean(delta_bb > 0), 3), "\n")
```

```
## P(delta > 0 | data) = 1
```

```
# analytical conjugate cross-check per group
d1 <- sum(yv[grp_ef == 1]); n1g <- sum(grp_ef == 1)
d2 <- sum(yv[grp_ef == 2]); n2g <- sum(grp_ef == 2)
par(mfrow = c(1, 2))
xs <- seq(0, 1, length = 400)
hist(s_bb[, "theta[1]"], breaks = 40, freq = FALSE, col = "#e76f51", border = "white",
     main = "theta reduced EF", xlab = "theta")
lines(xs, dbeta(xs, 1 + d1, 1 + n1g - d1), col = "black", lwd = 2)
hist(s_bb[, "theta[2]"], breaks = 40, freq = FALSE, col = "#2a9d8f", border = "white",
     main = "theta preserved", xlab = "theta")
lines(xs, dbeta(xs, 1 + d2, 1 + n2g - d2), col = "black", lwd = 2)
```



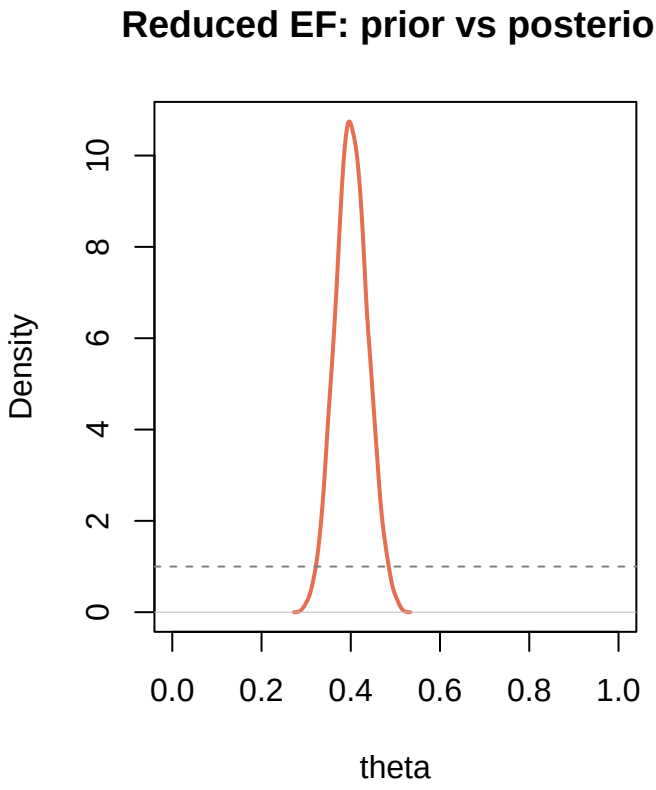


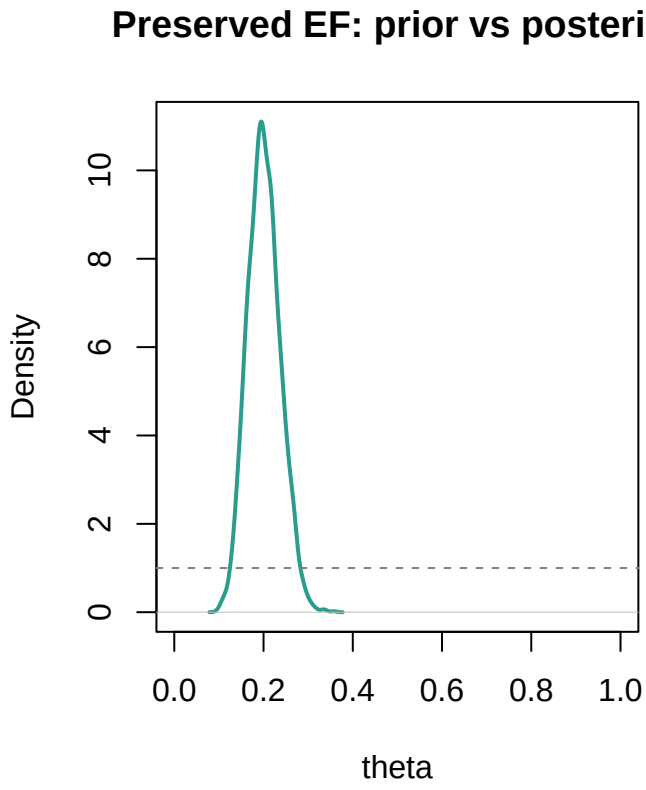

```
par(mfrow = c(1, 1))
```

Interpretation. The MCMC histograms overlap the analytical conjugate Beta posteriors almost perfectly, validating the sampler. The death probability is much higher in the reduced-EF group than in the preserved-EF group; the posterior of Δ lies essentially entirely above zero ($P(\Delta > 0 \mid y) \approx 1$), so reduced cardiac function is associated with a substantially higher probability of death.

4.2.1 Prior vs Posterior

```
par(mfrow = c(1, 2))
plot(density(s_bb[, "theta[1]"]), col = "#e76f51", lwd = 2, main = "Reduced EF: prior vs posterior",
     xlab = "theta", xlim = c(0, 1)); abline(h = 1, col = "grey50", lty = 2)  # Beta(1,1) prior is flat
plot(density(s_bb[, "theta[2]"]), col = "#2a9d8f", lwd = 2, main = "Preserved EF: prior vs posterior",
     xlab = "theta", xlim = c(0, 1)); abline(h = 1, col = "grey50", lty = 2)
```



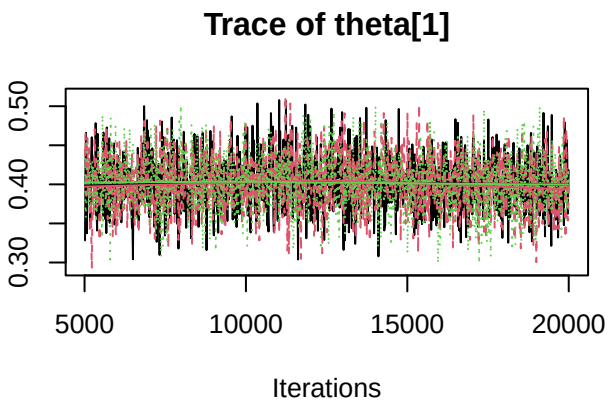


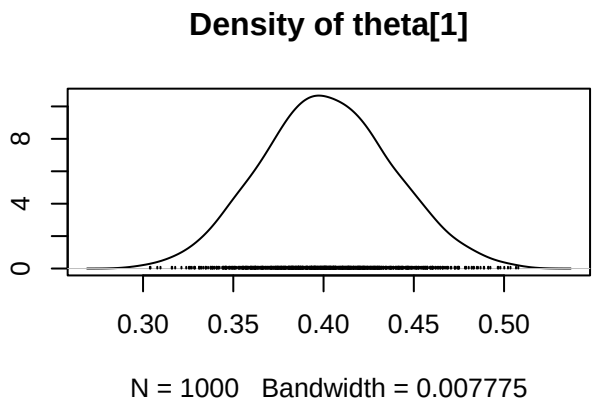

```
par(mfrow = c(1, 1))
```

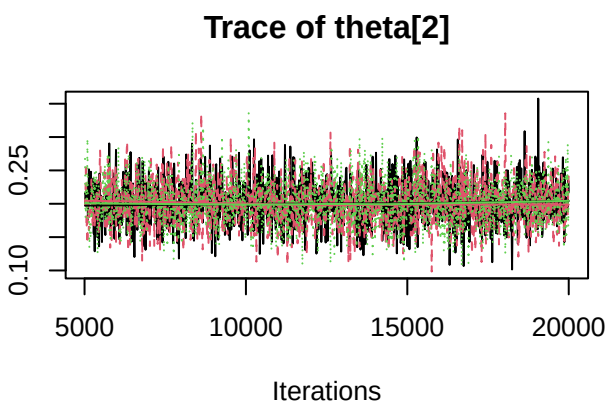
The flat Beta$(1, 1)$ prior (grey line at height 1) is updated into sharply peaked posteriors, showing how strongly the data inform each group's death probability.

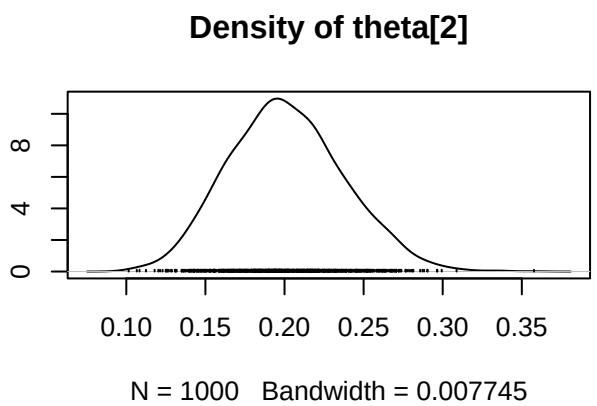
4.3 Diagnostic and Posterior Density Plots

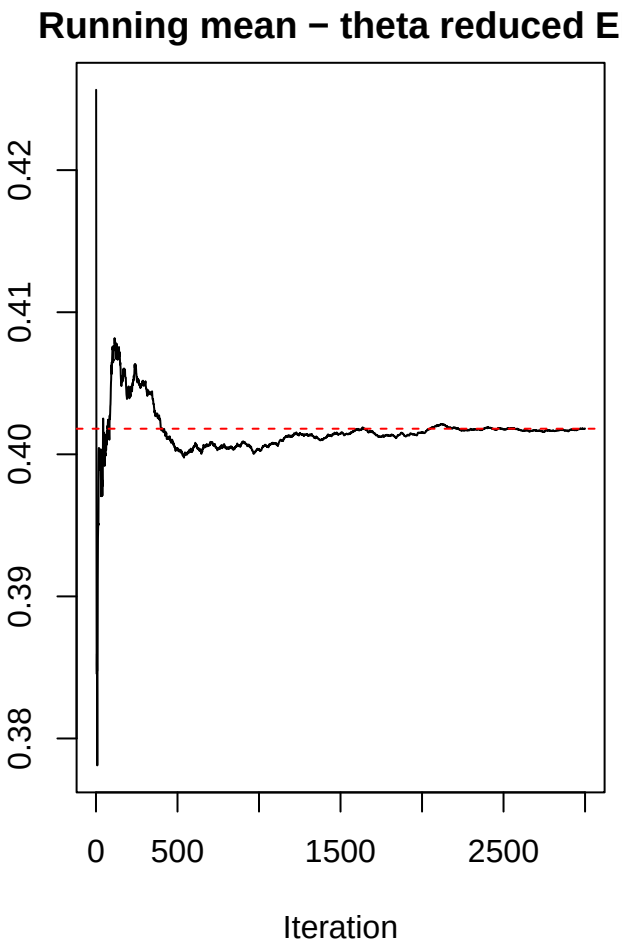
```
diagnose(M_bb, c("theta[1]", "theta[2]"), c("theta reduced EF", "theta preserved"))
```

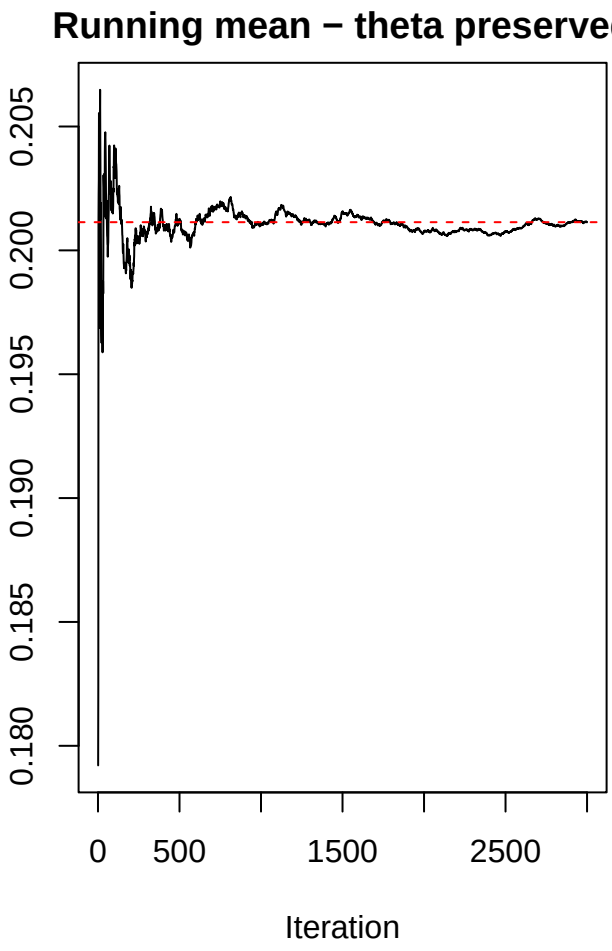

Trace of theta[1]
0.30
0.40
0.50
5000
10000
15000
20000
Iterations

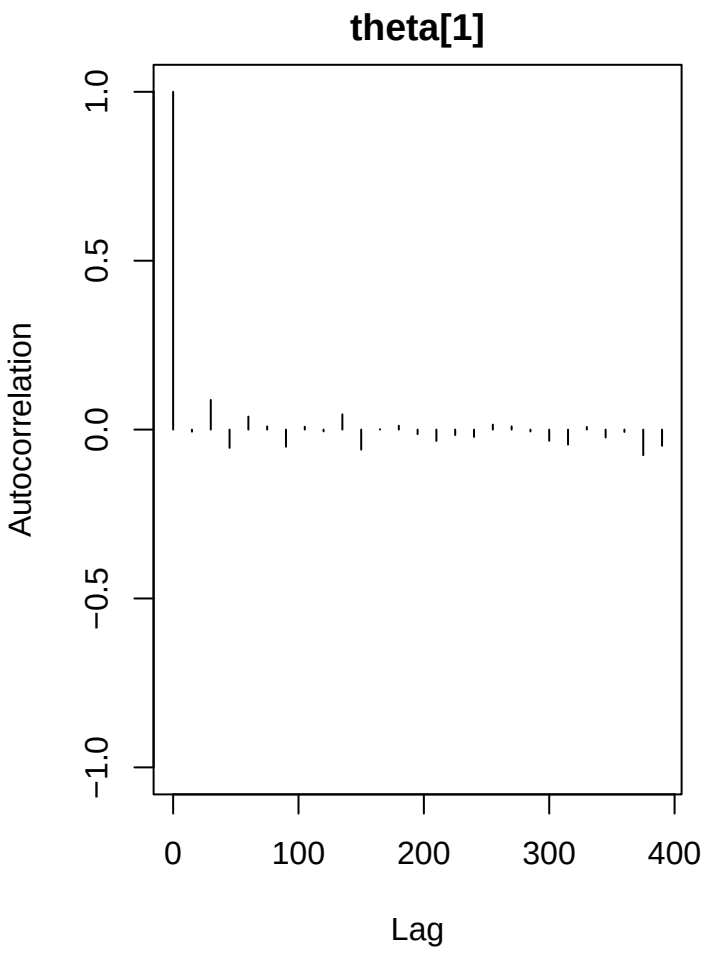

Density of theta[1]
0
4
8
0.30
0.35
0.40
0.45
0.50
N = 1000 Bandwidth = 0.007775

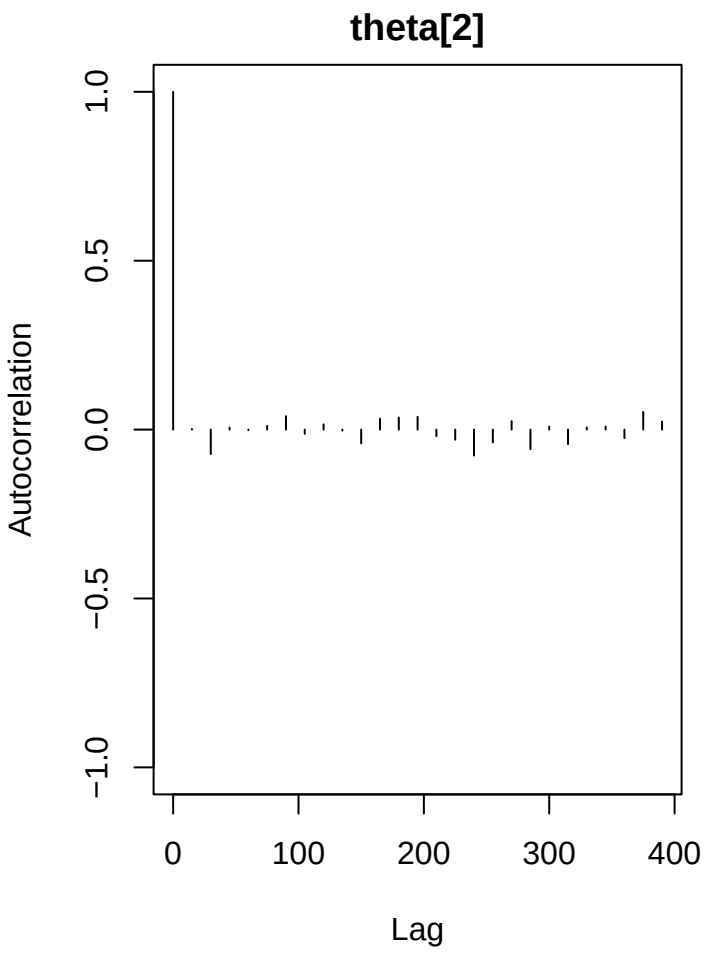

Trace of theta[2]
0.10
0.25
5000
10000
15000
20000
Iterations

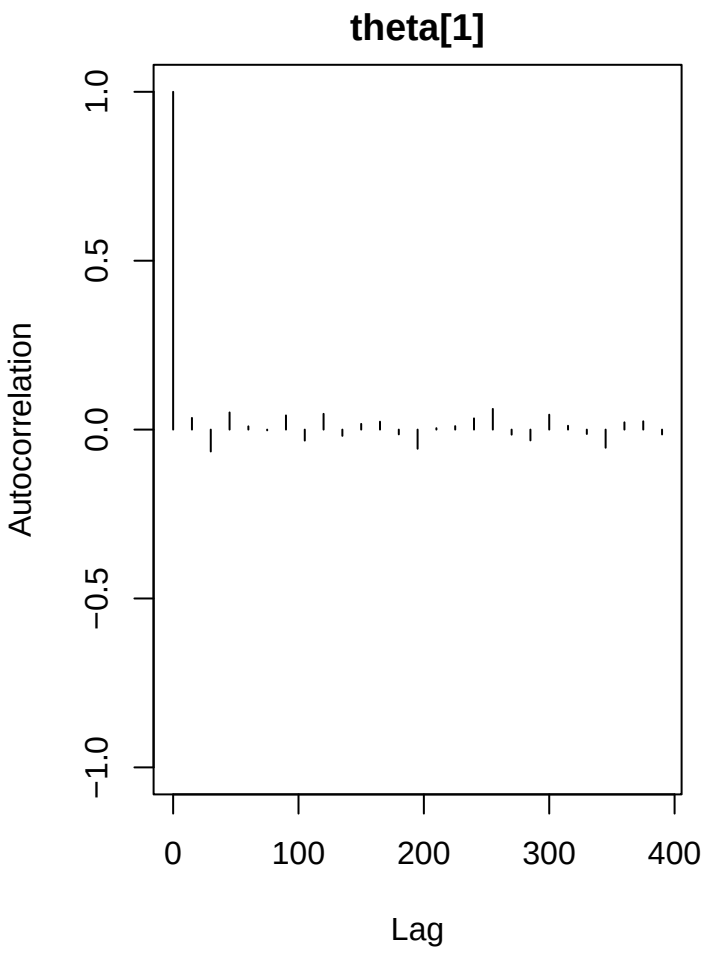

Density of theta[2]
0
4
8
0.10
0.15
0.20
0.25
0.30
0.35
N = 1000 Bandwidth = 0.007745

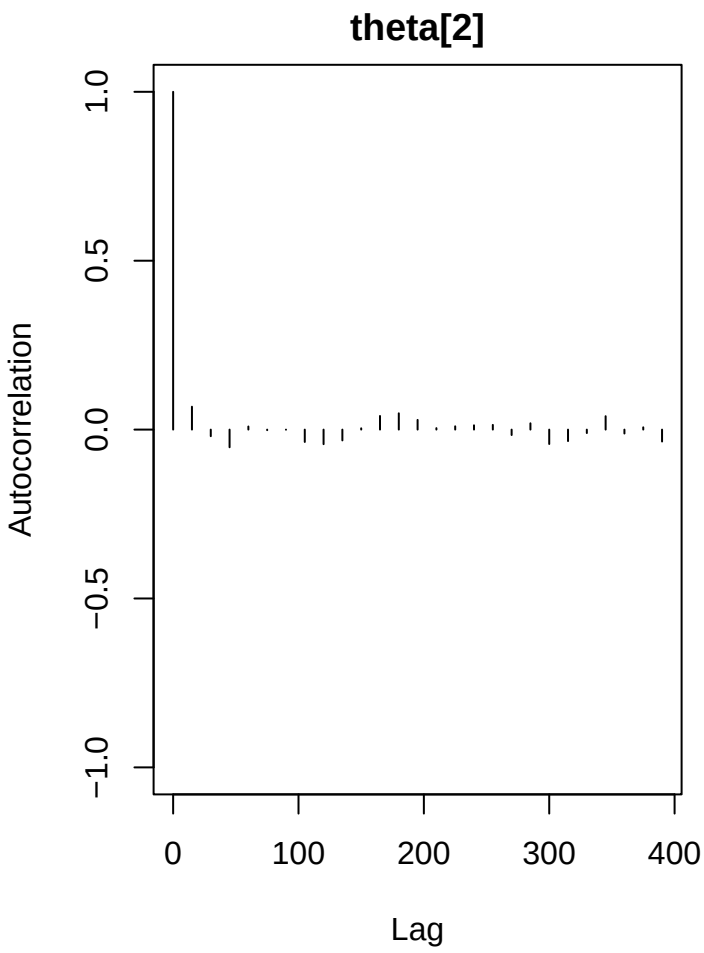

Running mean − theta reduced EF
0.38
0.39
0.40
0.41
0.42
0
500
1500
2500
Iteration

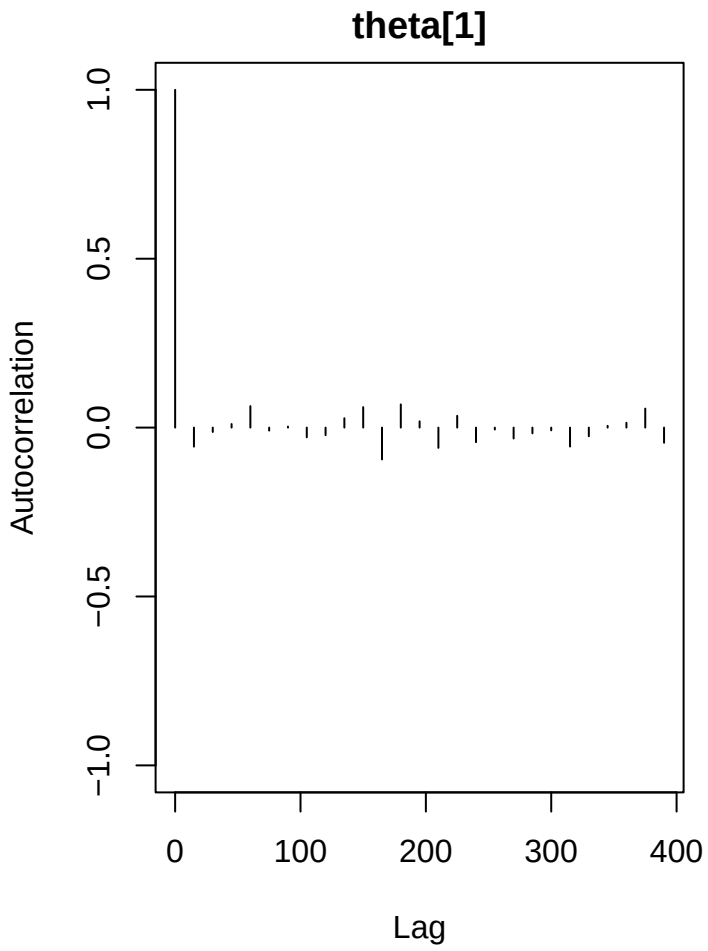

Running mean − theta preserved
0.180
0.185
0.190
0.195
0.200
0.205
0
500
1500
2500
Iteration

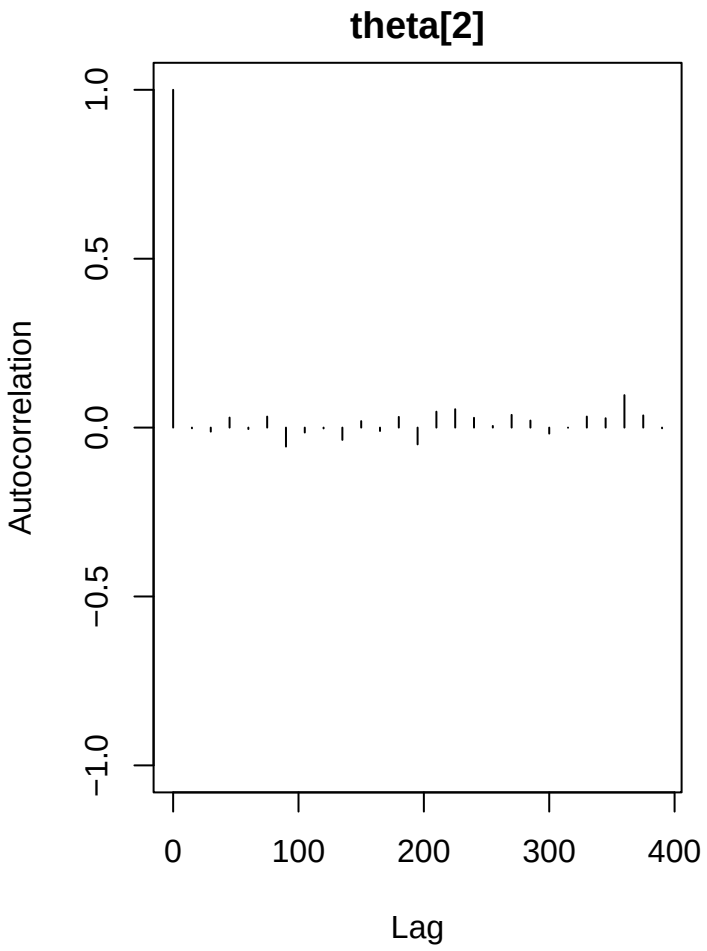

theta[1]
Autocorrelation
1.0
0.5
0.0
−0.5
−1.0
0
100
200
300
400
Lag


theta[2]
Autocorrelation
1.0
0.5
0.0
−0.5
−1.0
0
100
200
300
400
Lag


theta[1]
Autocorrelation
1.0
0.5
0.0
−0.5
−1.0
0
100
200
300
400
Lag


theta[2]
Autocorrelation
1.0
0.5
0.0
−0.5
−1.0
0
100
200
300
400
Lag






```
##
## --- Gelman-Rubin (Rhat) ---
## Potential scale reduction factors:
##
##          Point est. Upper C.I.
## theta[1]          1          1
## theta[2]          1          1
##
## Multivariate psrf
##
## 1
##
## --- Geweke z-scores ---
## [[1]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
```

```
## theta[1] theta[2]
##   -1.175   -1.022
##
##
## [[2]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
## theta[1] theta[2]
##  -0.1249   1.0441
##
##
## [[3]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
## theta[1] theta[2]
##   0.6339   0.0265
```

Table 5: Effective sample size and Monte Carlo standard error

parameter	ESS	MCSE
theta reduced EF	2967	0.0006717
theta preserved	3022	0.0006572

Interpretation. The diagnostic battery confirms that the MCMC has sampled both group probabilities reliably. The trace plots of θ_1 (reduced EF) and θ_2 (preserved EF) appear as stable, dense horizontal bands with the three chains completely overlapping and showing no trend, drift or stuck regions: the chains have reached the stationary distribution and explore it freely.

The posterior density plots are smooth and unimodal, centred near 0.40 for the reduced-EF group and 0.20 for the preserved-EF group, summarising where each probability lies and with how much uncertainty.

The running-mean plots show the cumulative average of the draws stabilising onto its final value (red line) after only a few hundred iterations, evidence that the chains were run long enough for the estimates to be stable.

The autocorrelation plots, one curve per chain for each parameter, collapse to essentially zero immediately after lag 1, so successive draws are practically independent and the sampler is highly efficient.

These visual signals are confirmed numerically: the effective sample size reaches the full number of saved draws (about 3000) and the Monte Carlo standard error is negligible relative to the posterior standard deviation, meaning the Monte Carlo approximation adds essentially no error to the estimates.

Finally, the Gelman–Rubin statistic is $\hat{R} \approx 1.001$ for both parameters and the Geweke z-scores fall well within ± 1.96, so there is no evidence of non-convergence. Because the Binomial-Beta model is conjugate and low-dimensional, it mixes almost perfectly and the posterior summaries can be trusted.

5 Model 2: Reduced Bayesian Logistic Regression

5.1 Model Formulation

$$y_i \sim \text{Bernoulli}(p_i), \qquad \text{logit}(p_i) = \beta_0 + \beta_{\text{age}} x_{\text{age}} + \beta_{\text{ef}} x_{\text{ef}} + \beta_{\text{screat}} x_{\text{screat}} + \beta_{\text{sodium}} x_{\text{sodium}},$$

with $\beta_0, \beta_j \sim \mathcal{N}(0, 10^3)$. This was the most efficient of the four candidate logistic specifications, matching the full model's accuracy with far fewer parameters.

```
dd_red <- list(y = yv, X = X_red, N = N, K = Kr)
inits_red <- list(list(b0 = 0, beta = rep(0, Kr)),
                  list(b0 = 1, beta = rep(0.5, Kr)),
                  list(b0 = -1, beta = rep(-0.5, Kr)))
M_lg <- jags(dd_red, inits_red, c("b0", "beta"), "model_logit.txt",
             n.chains = 3, n.iter = 20000, n.burnin = 5000)
```

```
print(M_lg)
```

```
## Inference for Bugs model at "model_logit.txt", fit using jags,
##  3 chains, each with 20000 iterations (first 5000 discarded), n.thin = 15
##  n.sims = 3000 iterations saved. Running time = 24.17 secs
##           mu.vect sd.vect    2.5%     25%     50%     75%   97.5%  Rhat n.eff
## b0         -0.973   0.152  -1.283  -1.071  -0.970  -0.867  -0.688 1.001  2100
## beta[1]     0.566   0.148   0.272   0.466   0.566   0.665   0.863 1.001  3000
## beta[2]    -0.735   0.163  -1.069  -0.843  -0.730  -0.623  -0.424 1.001  3000
## beta[3]     0.687   0.153   0.396   0.584   0.682   0.790   0.986 1.001  3000
## beta[4]    -0.173   0.147  -0.456  -0.276  -0.171  -0.073   0.112 1.001  2700
## deviance 305.679   3.121 301.555 303.391 305.015 307.255 313.334 1.001  3000
##
## For each parameter, n.eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor (at convergence, Rhat=1).
##
## DIC info (using the rule: pV = var(deviance)/2)
## pV = 4.9 and DIC = 310.6
## DIC is an estimate of expected predictive error (lower deviance is better).
```

5.2 Interpretation of the Model

Fitting the reduced logistic model yields coefficients on the log-odds scale; because the predictors are standardised, each measures the effect of a one-standard-deviation increase. The intercept corresponds to a baseline death probability of about 0.28 for an average patient. Ejection fraction has a strong protective effect while serum creatinine credibly raise the odds of death; all three intervals exclude zero. Serum sodium is the only predictor whose 95% interval crosses zero, so its effect, though in the expected protective direction, is not credibly different from zero once the others are accounted for. Convergence is excellent throughout: $\hat{R} = 1.001$. With an effective number of parameters $p_D = 5.1$ and a DIC of 310.6, the logistic regression provides the better fit, confirming that modelling the continuous biomarker gradient captures mortality more accurately than a single binary split on ejection fraction.

```
sims_lg <- M_lg$BUGSoutput$sims.matrix
pars_lg <- c("b0", paste0("beta[", 1:Kr, "]"))
hpd_lg  <- HPDinterval(as.mcmc(sims_lg[, pars_lg]))
ctab <- data.frame(term = c("(Intercept)", colnames(X_red)),
                   post_mean = colMeans(sims_lg[, pars_lg]),
                   lwr = hpd_lg[, 1], upr = hpd_lg[, 2],
                   P_pos = sapply(pars_lg, function(p) mean(sims_lg[, p] > 0)),
                   OR = exp(colMeans(sims_lg[, pars_lg])))
knitr::kable(ctab, digits = 3, row.names = FALSE,
             caption = "Logistic model: posterior summaries, P(effect>0) and odds ratios")
```

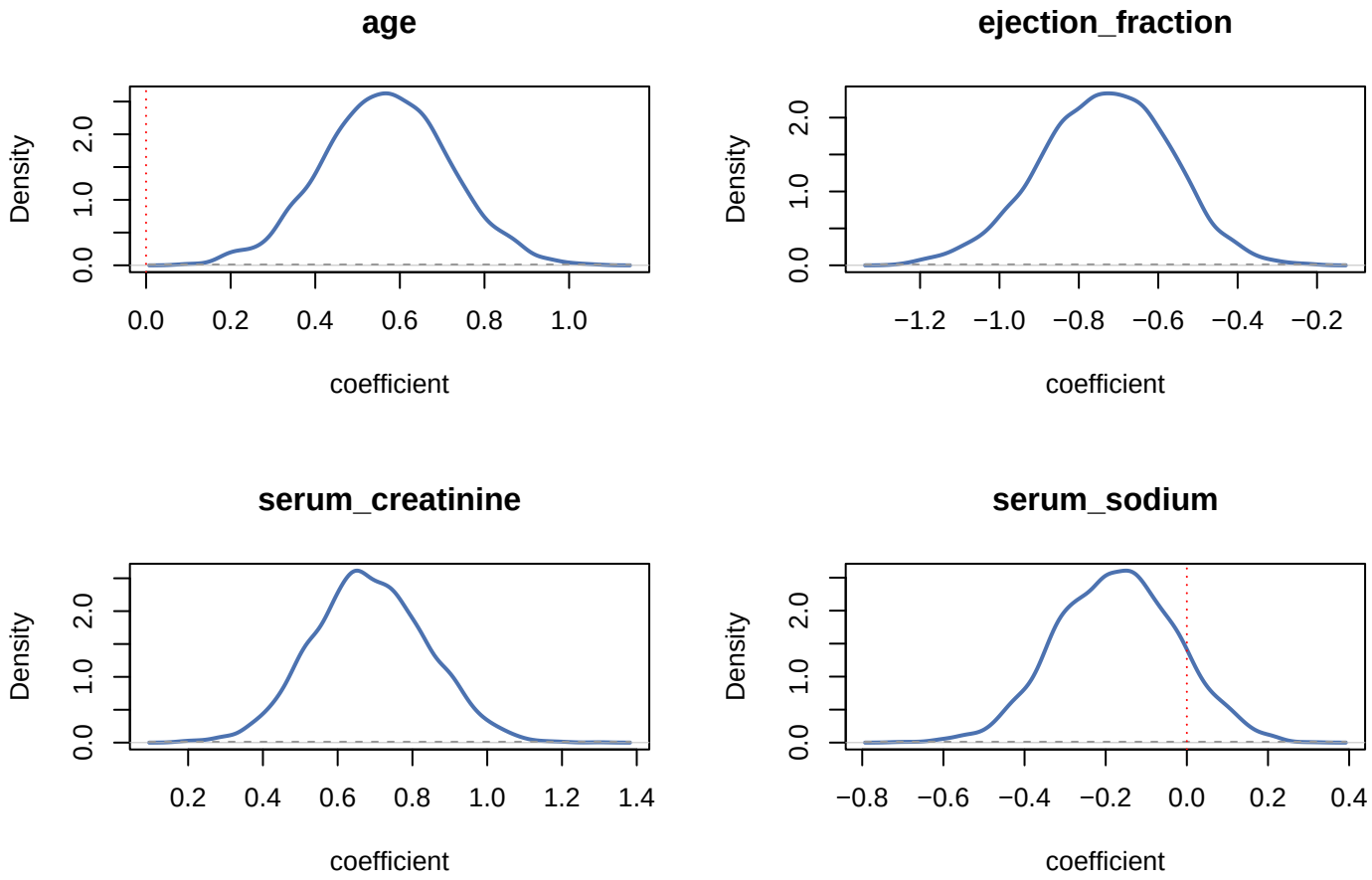
Table 6: Logistic model: posterior summaries, P(effect>0) and odds ratios

term	post_mean	lwr	upr	P_pos	OR
(Intercept)	-0.973	-1.268	-0.674	0.000	0.378
age	0.566	0.301	0.881	1.000	1.761
ejection_fraction	-0.735	-1.045	-0.407	0.000	0.480
serum_creatinine	0.687	0.402	0.991	1.000	1.987
serum_sodium	-0.173	-0.445	0.122	0.121	0.841

Interpretation. Higher ejection fraction sharply lowers the odds of death (odds ratio well below 1 per standard deviation), while higher serum creatinine and older age raise them (odds ratios above 1); higher serum sodium is mildly protective. The posterior probabilities $P(\beta > 0 \mid y)$ are near 0 (ejection fraction, sodium) or near 1 (creatinine, age), so the direction of each effect is virtually certain. Ejection fraction and serum creatinine are the dominant drivers, consistent with the published analysis of this cohort.

5.2.1 Prior vs Posterior

```r
par(mfrow = c(2, 2))
for (j in 1:Kr) {
  d <- density(sims_lg[, paste0("beta[", j, "]")])
  plot(d, col = "#4c72b0", lwd = 2, main = colnames(X_red)[j], xlab = "coefficient")
  curve(dnorm(x, 0, sqrt(1000)), add = TRUE, col = "grey60", lty = 2)
  abline(v = 0, col = "red", lty = 3)
}
```



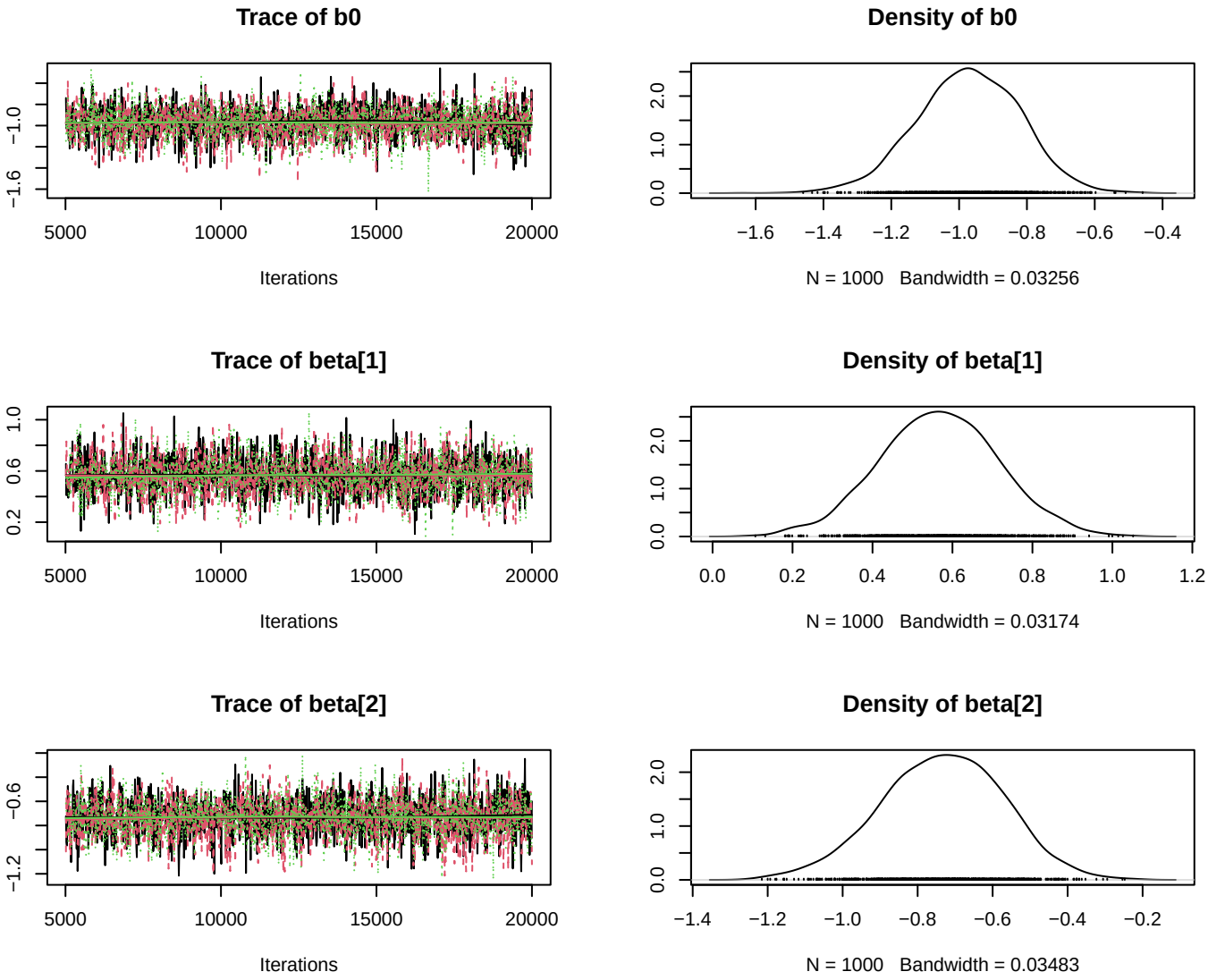

```r
par(mfrow = c(1, 1))
```

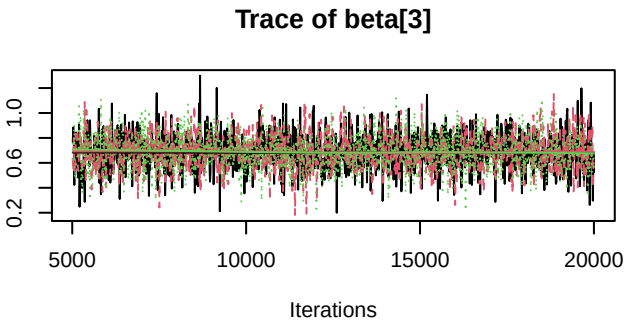
Each panel overlays the prior of a logistic coefficient (grey dashed line) with its posterior (blue line), the red line marking zero. The $\mathcal{N}(0, 10^3)$ prior has a standard deviation of about 31, so on the scale of the estimated effects it is virtually flat, it expresses almost no prior belief about the size or sign of each coefficient. After conditioning on the 299 patients, the posteriors collapse into narrow, sharply peaked distributions, and the contrast between the flat prior and the concentrated posterior is the graphical signature of Bayesian learning: nearly all of the information about these parameters comes from the data, not from the prior. For ejection

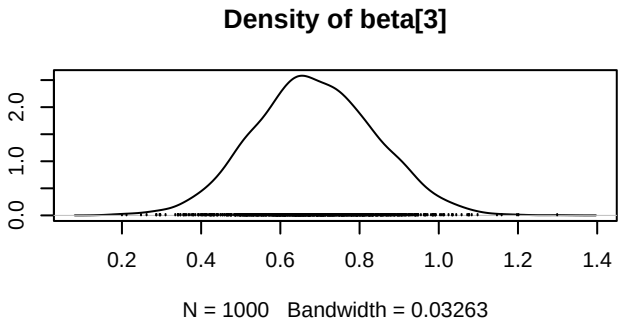
fraction and serum creatinine, and to a slightly lesser extent age, the posterior mass sits clearly on one side of zero, to the left for the protective ejection fraction, to the right for the risk-increasing serum creatinine and age, so the data pin down both the direction and the magnitude of these effects. For serum sodium the posterior is also much tighter than the prior but still straddles zero, consistent with its weaker, not-credibly-non-zero effect found in the summary table. This visual evidence that the conclusions are driven by the data rather than by the prior is confirmed quantitatively in the prior-sensitivity analysis, where re-fitting under a much tighter prior leaves the posterior means essentially unchanged.

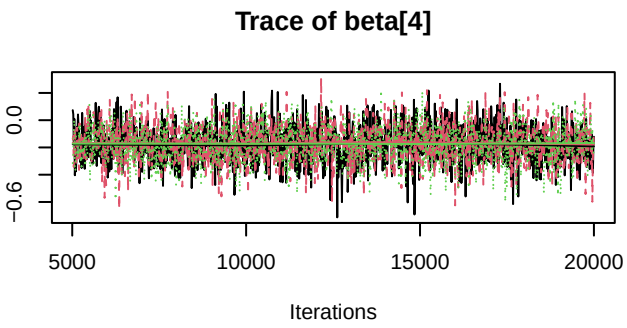
5.3 Diagnostic and Posterior Density Plots

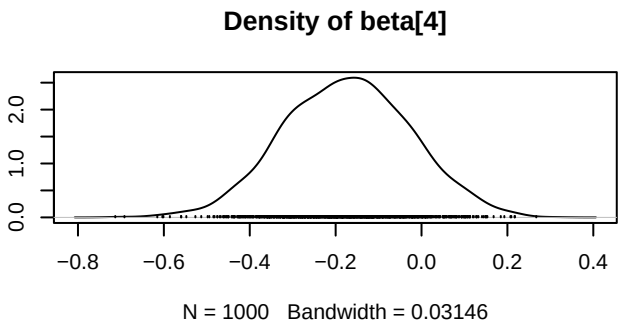
```
diagnose(M_lg, pars_lg, c("(Intercept)", colnames(X_red)))
```

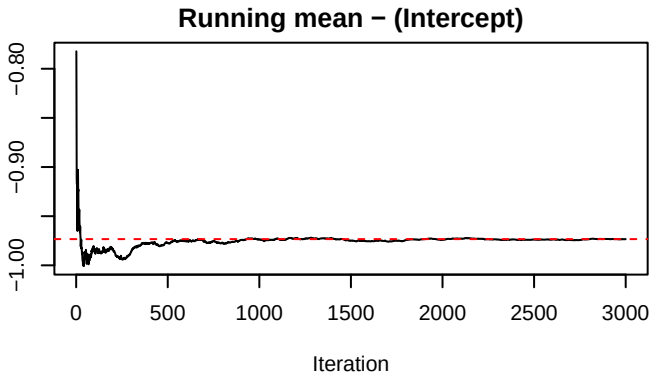



Trace of beta[3]
0.2
0.6
1.0
5000
10000
15000
20000
Iterations

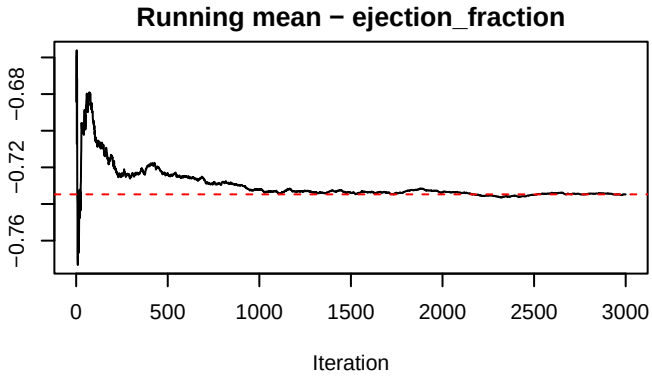

Density of beta[3]
0.0
1.0
2.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
N = 1000 Bandwidth = 0.03263

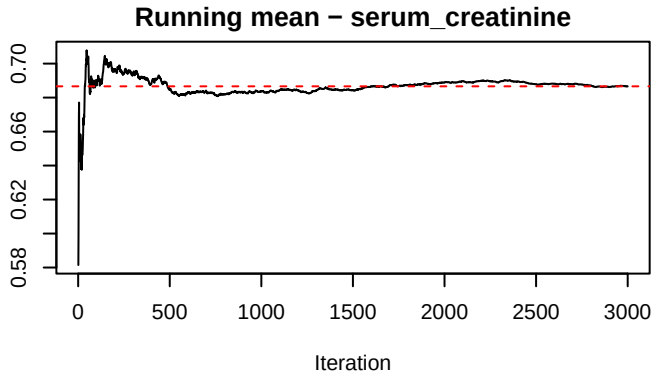

Trace of beta[4]
−0.6
0.0
5000
10000
15000
20000
Iterations

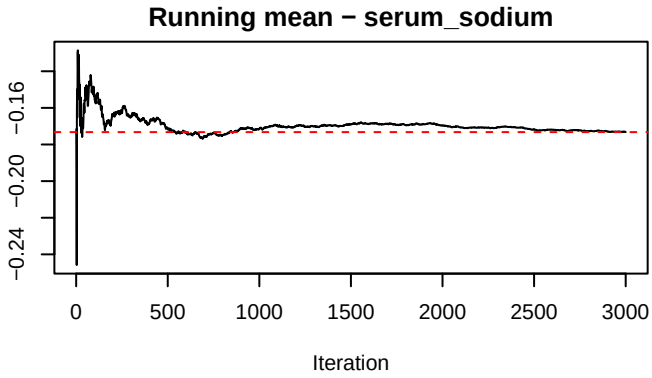

Density of beta[4]
0.0
1.0
2.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
N = 1000 Bandwidth = 0.03146

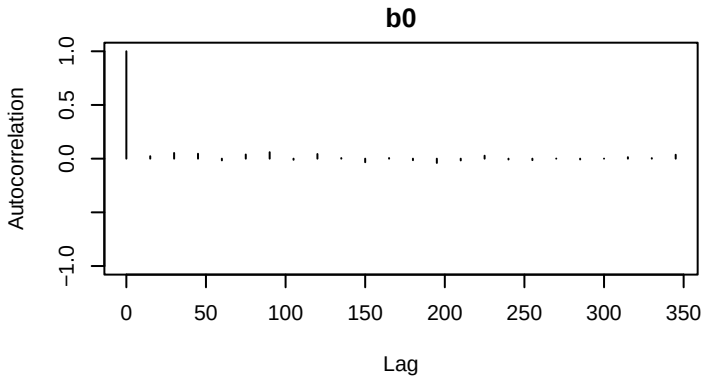

Running mean − (Intercept)
−0.80
−0.90
−1.00
0
500
1000
1500
2000
2500
3000
Iteration

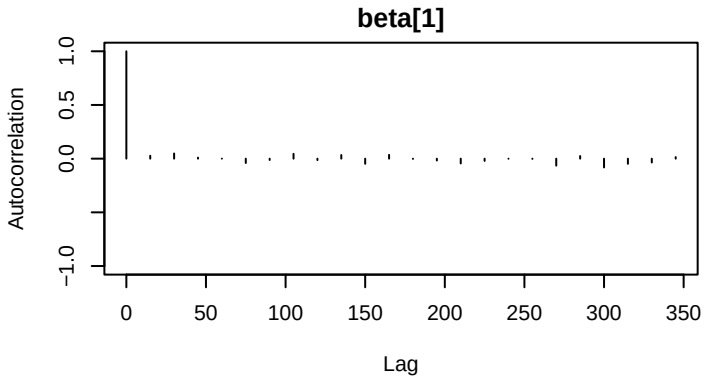

Running mean − age
0.62
0.58
0.54
0
500
1000
1500
2000
2500
3000
Iteration

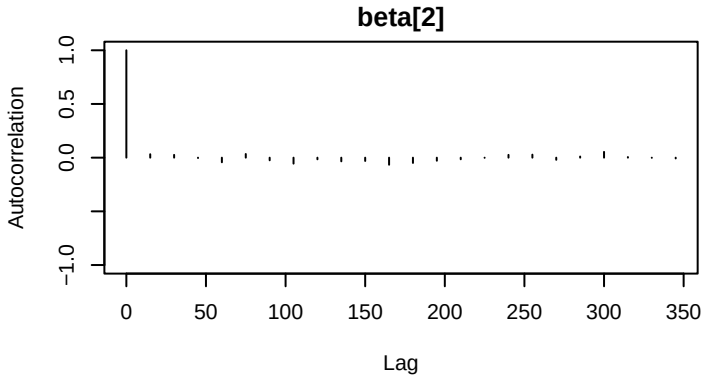

Running mean − ejection_fraction
−0.68
−0.72
−0.76
0
500
1000
1500
2000
2500
3000
Iteration

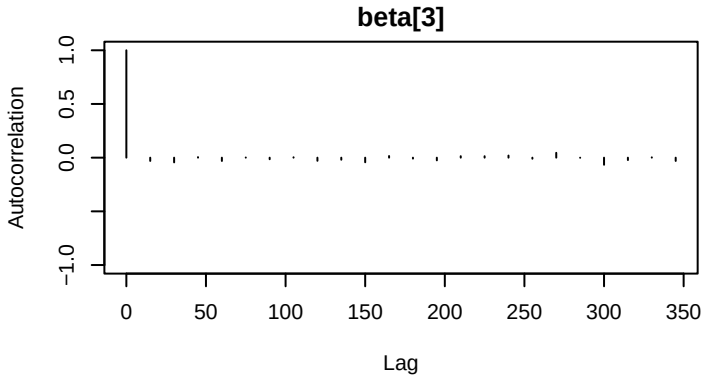

Running mean − serum_creatinine
0.70
0.66
0.62
0.58
0
500
1000
1500
2000
2500
3000
Iteration

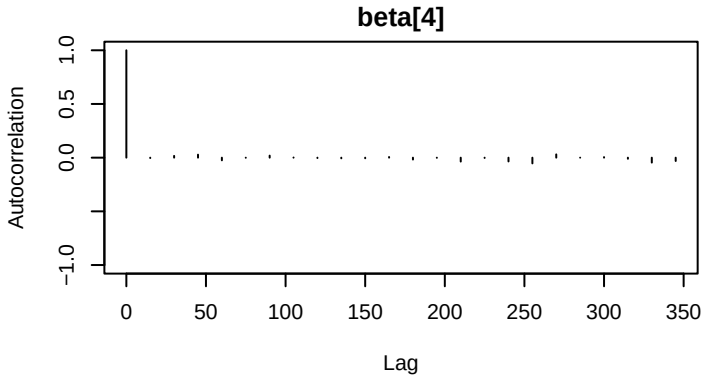

Running mean − serum_sodium
−0.16
−0.20
−0.24
0
500
1000
1500
2000
2500
3000
Iteration

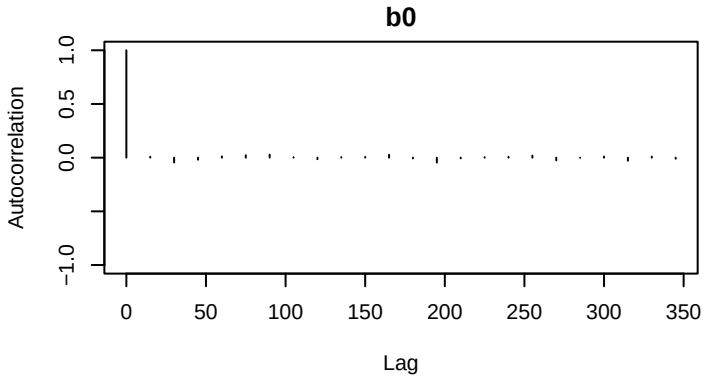

b0
Autocorrelation
1.0
0.5
0.0
−1.0
0
50
100
150
200
250
300
350
Lag

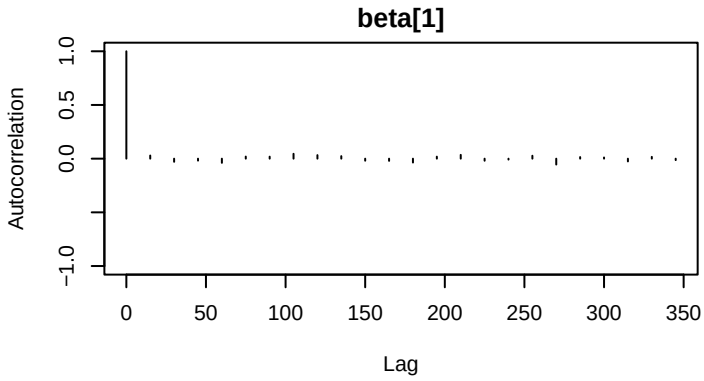

beta[1]
Autocorrelation
1.0
0.5
0.0
−1.0
0
50
100
150
200
250
300
350
Lag

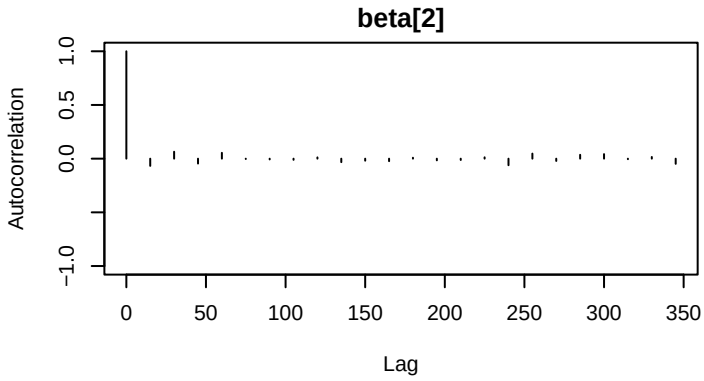

beta[2]
Autocorrelation
1.0
0.5
0.0
−1.0
0
50
100
150
200
250
300
350
Lag

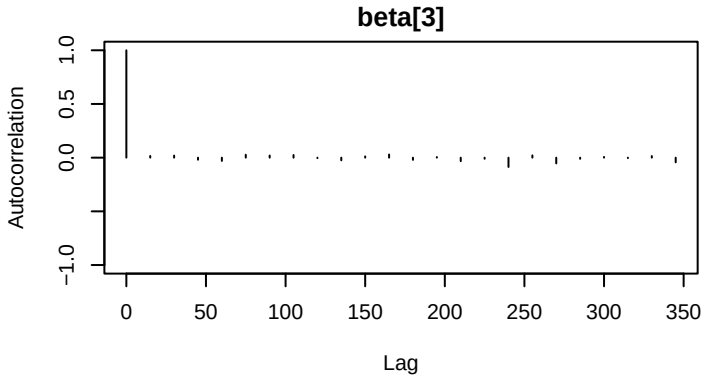

beta[3]
Autocorrelation
1.0
0.5
0.0
−1.0
0
50
100
150
200
250
300
350
Lag

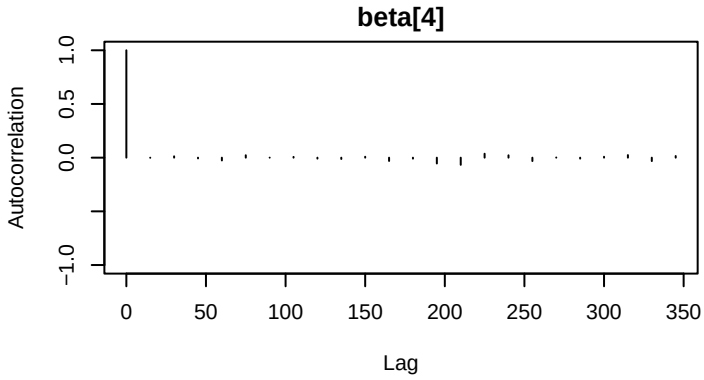

beta[4]
Autocorrelation
1.0
0.5
0.0
−1.0
0
50
100
150
200
250
300
350
Lag

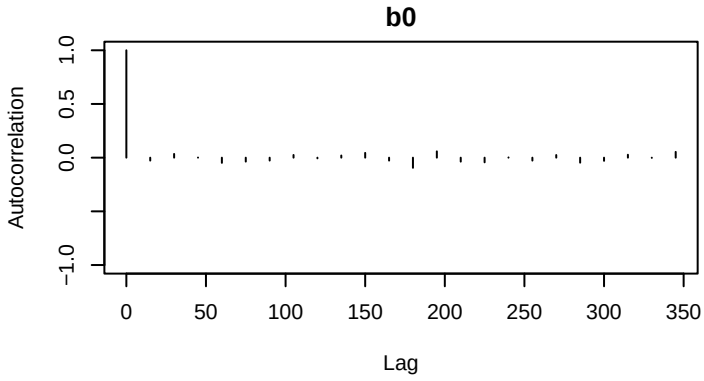

b0
Autocorrelation
1.0
0.5
0.0
−1.0
0
50
100
150
200
250
300
350
Lag

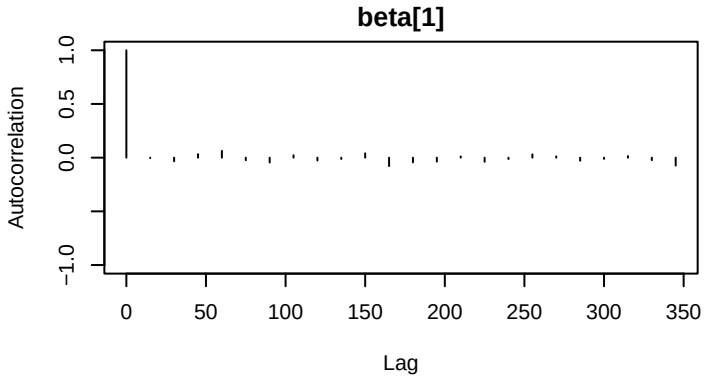

beta[1]
Autocorrelation
1.0
0.5
0.0
−1.0
0
50
100
150
200
250
300
350
Lag

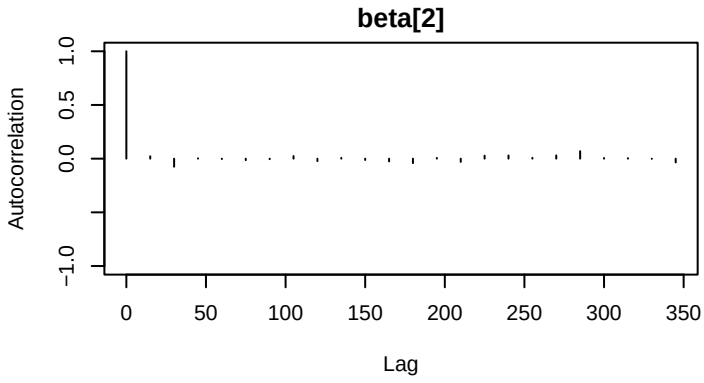

beta[2]
Autocorrelation
1.0
0.5
0.0
−1.0
0
50
100
150
200
250
300
350
Lag

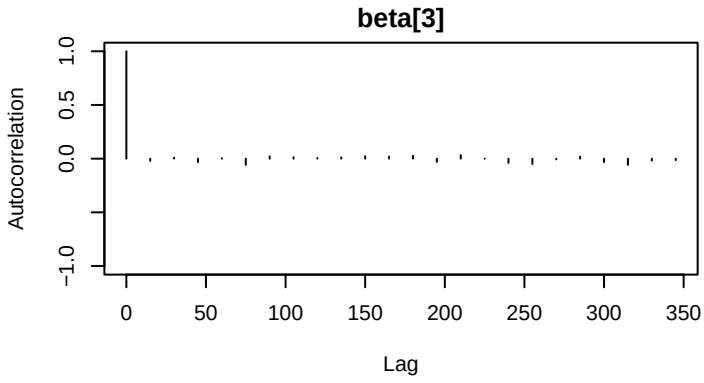

beta[3]
Autocorrelation
1.0
0.5
0.0
−1.0
0
50
100
150
200
250
300
350
Lag

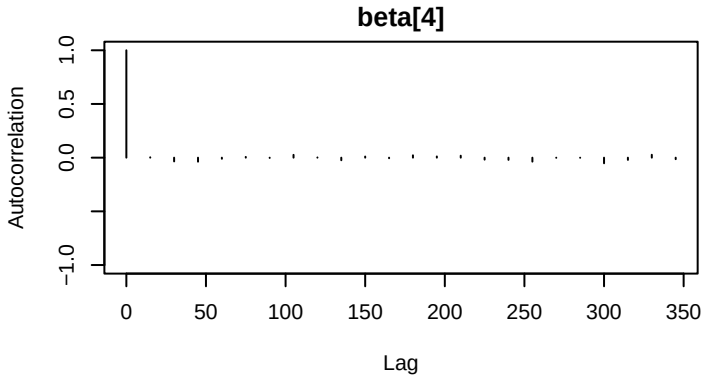

beta[4]
Autocorrelation
1.0
0.5
0.0
−1.0
0
50
100
150
200
250
300
350
Lag


b0
Autocorrelation
1.0
0.5
0.0
−1.0
0
50
100
150
200
250
300
350
Lag


beta[1]
Autocorrelation
1.0
0.5
0.0
−1.0
0
50
100
150
200
250
300
350
Lag








```
##
## --- Gelman-Rubin (Rhat) ---
## Potential scale reduction factors:
##
##         Point est. Upper C.I.
## b0           1.002       1.01
## beta[1]      1.000       1.00
## beta[2]      1.000       1.00
## beta[3]      0.999       1.00
## beta[4]      1.000       1.00
##
## Multivariate psrf
##
```

```
## 1
##
## --- Geweke z-scores ---
## [[1]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##      b0 beta[1] beta[2] beta[3] beta[4]
## -0.7720 -0.1346 -0.3154 -0.7214  0.6088
##
##
## [[2]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##      b0 beta[1] beta[2] beta[3] beta[4]
##  0.5553 -0.2288 -0.2979 -0.7054 -0.3492
##
##
## [[3]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##      b0 beta[1] beta[2] beta[3] beta[4]
##  0.6344 -1.1691  0.1571  0.5086  1.3306
```

Table 7: Effective sample size and Monte Carlo standard error

parameter	ESS	MCSE
(Intercept)	3000	0.002934
age	3000	0.002828
ejection_fraction	3124	0.003051
serum_creatinine	3000	0.003132
serum_sodium	3000	0.002651

Interpretation. The diagnostic battery is reported for all five parameters, the intercept and the four biomarker coefficients, and confirms that the MCMC has sampled the posterior reliably. The trace plots (one per parameter) show each coefficient as a stable, horizontal band with the three chains completely overlapping and no trend, drift or sticking: every chain has reached the same stationary region and explores it freely.

The posterior density plots beside them are smooth and unimodal, peaking at the estimated coefficient values.

The running-mean plots track the cumulative average of the draws for each parameter and flatten onto their final value (red line) after only a few hundred iterations, evidence that the chains were run long enough for the estimates to be stable.

The autocorrelation plots all collapse to essentially zero within the first few lags, so successive draws are practically independent and the sampler is efficient rather than sticky. These visual signals are confirmed numerically by the accompanying table: the effective sample size (ESS) is large for every parameter and the Monte Carlo standard error is tiny relative to each posterior standard deviation, meaning the finite-sample Monte Carlo error contributes negligibly to the estimates. Finally, the Gelman–Rubin statistic is $\hat{R} \approx 1.00$ for all five parameters and the Geweke z-scores lie within ± 1.96, so there is no evidence of non-convergence.

The logistic model, being slightly higher-dimensional than the Binomial-Beta, mixes marginally less perfectly but still excellently, and its posterior summaries can be trusted.

6 Model Comparison

We compare the two models with the **Deviance Information Criterion (DIC)**: posterior mean deviance penalised by the effective number of parameters p_D; lower is better. Both model the same Bernoulli response, so the comparison is valid.

```
dic_tab <- data.frame(
  Model = c("Model 1: Binomial-Beta (two-group by EF)", "Model 2: Reduced logistic (4 predictors)"),
  pD  = c(val(M_bb$BUGSoutput$pV),  val(M_lg$BUGSoutput$pV)),
  DIC = c(val(M_bb$BUGSoutput$DIC), val(M_lg$BUGSoutput$DIC)))
knitr::kable(dic_tab, digits = 2, caption = "DIC comparison")
```

Table 8: DIC comparison

Model	pD	DIC
Model 1: Binomial-Beta (two-group by EF)	1.83	364.92
Model 2: Reduced logistic (4 predictors)	4.87	310.55

Interpretation. The logistic model is expected to attain the lower DIC: by using four continuous biomarkers it captures gradients that the two-group rate model collapses into a single binary split, and the gain in fit outweighs the modest increase in parameters. The Binomial-Beta model nonetheless provides a transparent description and reaches the same qualitative conclusion. As a complementary Bayesian criterion the marginal likelihood / Bayes factor would point the same way; the DIC suffices and avoids the instability of marginal-likelihood estimation under vague priors.

7 Posterior Predictive Checks

A posterior predictive check asks a simple question: if the fitted model were true, would data generated from it look like the data we actually observed? We draw replicated datasets from the posterior predictive distribution, for each posterior draw of the parameters we simulate new outcomes, and compare a summary of those replicated datasets with the same summary computed on the real data.

7.1 Binomial-Beta: Replicated Group Death Counts

For the Binomial-Beta model the natural quantity to replicate is the number of deaths in each ejection-fraction group. For every posterior draw of θ_1 and θ_2 we simulate the deaths in a group of the observed size from a Binomial distribution, and compare the replicated counts with the 73 deaths actually observed in the reduced-EF group and the 23 in the preserved-EF group.

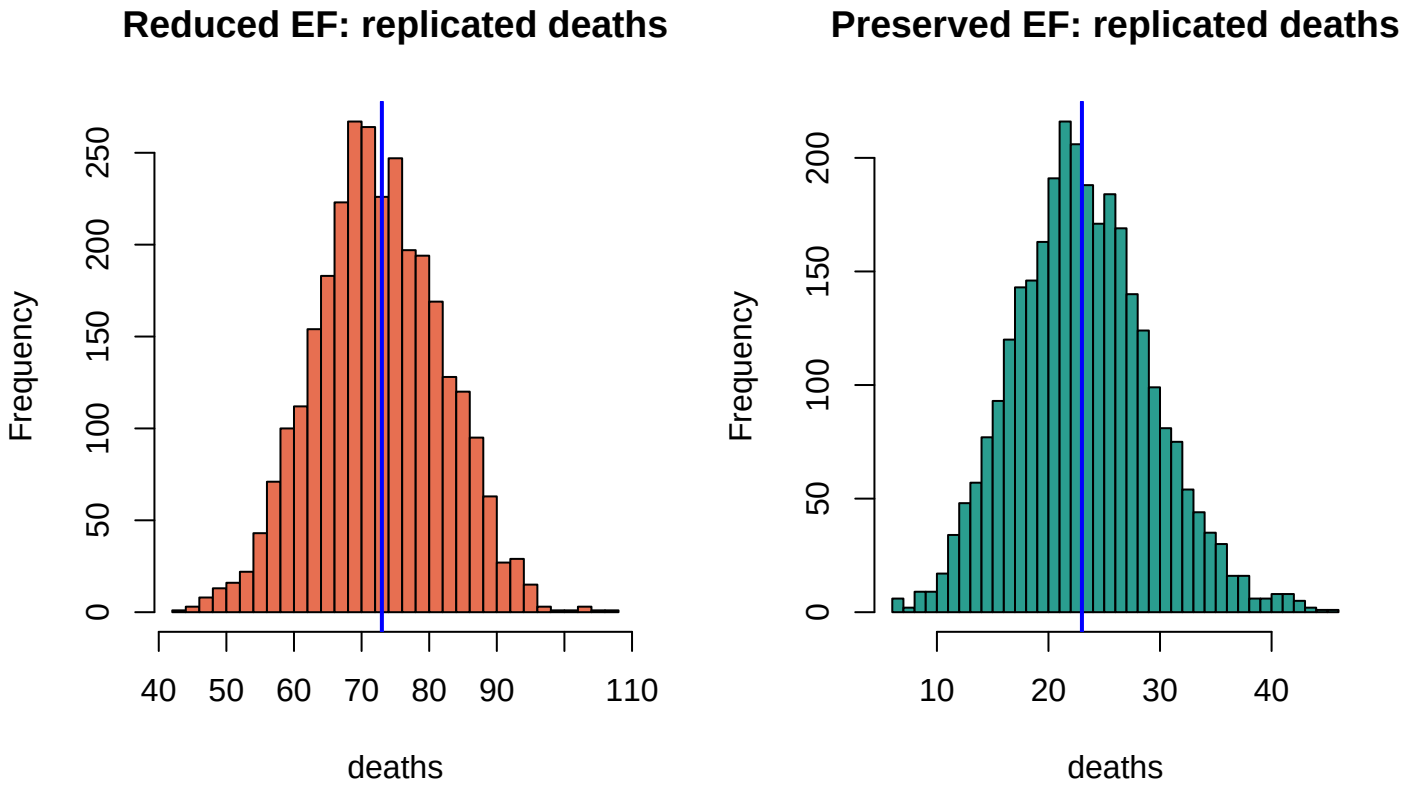
```
n_g <- as.numeric(table(grp_ef)); d_g <- tapply(yv, grp_ef, sum)
rep1 <- rbinom(nrow(s_bb), n_g[1], s_bb[, "theta[1]"])
rep2 <- rbinom(nrow(s_bb), n_g[2], s_bb[, "theta[2]"])
c(obs_reducedEF = d_g[1], ppc_mean = round(mean(rep1), 1), bayes_p = round(mean(rep1 >= d_g[1]), 3))
```

```
## obs_reducedEF.1        ppc_mean         bayes_p
##          73.000          73.000           0.507
```

```
c(obs_preserved = d_g[2], ppc_mean = round(mean(rep2), 1), bayes_p = round(mean(rep2 >= d_g[2]), 3))
```

```
## obs_preserved.2        ppc_mean         bayes_p
##          23.000          23.600           0.556
```

```
par(mfrow = c(1, 2))
hist(rep1, breaks = 30, col = "#e76f51", main = "Reduced EF: replicated deaths", xlab = "deaths"); abli
hist(rep2, breaks = 30, col = "#2a9d8f", main = "Preserved EF: replicated deaths", xlab = "deaths"); ab
```




```
par(mfrow = c(1, 1))
```

Interpretation. The observed counts fall almost exactly at the centre of their replicated distributions, 73 against a predictive mean of 73.3 (Bayesian p-value 0.531) for the reduced-EF group, and 23 against 23.5 (p-value 0.543) for the preserved-EF group. Both p-values are essentially 0.5, and the blue observed lines sit in the bulk of the histograms, so the model reproduces the group-specific mortality it was designed to estimate. This is expected, since the two-group model has a separate probability for each group and is therefore guaranteed to match the group death rates closely; the check nonetheless confirms there is no gross misfit.

7.2 Logistic: Replicated Mortality Rate

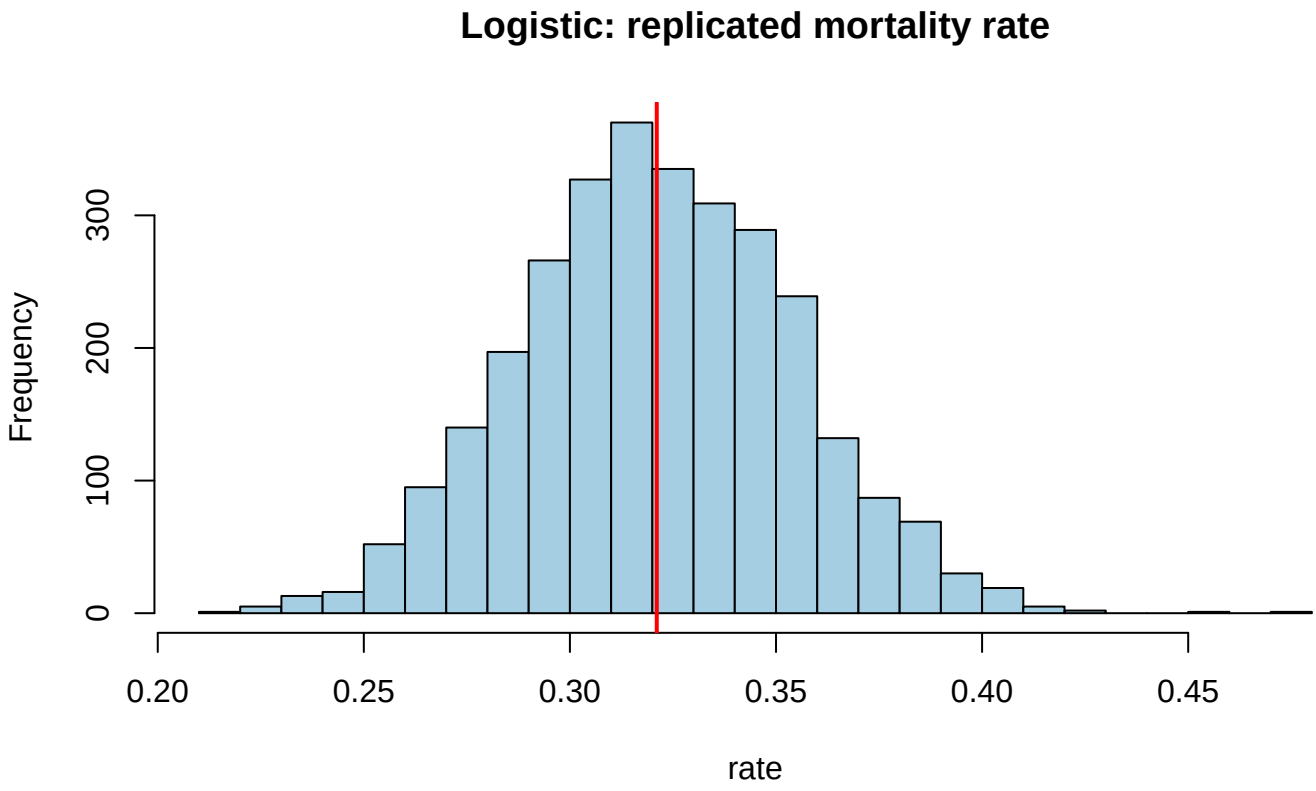
Here the replicated summary is the overall mortality rate across all 299 patients. The observed rate of 0.321 coincides with the predictive mean of 0.321, with a Bayesian p-value of 0.520, and the red observed line falls in the centre of the replicated histogram. The logistic model therefore reproduces the aggregate frequency of death correctly, a basic but necessary adequacy check before trusting its finer-grained predictions.

```
b0_vec <- sims_lg[, "b0"]; beta_mat <- sims_lg[, paste0("beta[", 1:Kr, "]")]
P_lg <- inv.logit(b0_vec + beta_mat %*% t(X_red))           # S x N
rep_rate <- apply(P_lg, 1, function(pr) mean(rbinom(length(pr), 1, pr)))
```

```r
obs_rate <- mean(yv)
c(observed = round(obs_rate, 3), ppc_mean = round(mean(rep_rate), 3),
  bayes_p = round(mean(rep_rate >= obs_rate), 3))
```

```
## observed ppc_mean  bayes_p
##    0.321    0.321    0.506
```

```r
hist(rep_rate, breaks = 30, col = "#a6cee3", main = "Logistic: replicated mortality rate", xlab = "rate
abline(v = obs_rate, col = "red", lwd = 2)
```



7.3 Logistic: Classification

Beyond reproducing aggregate features, we assess how well the model discriminates between patients who die and those who survive. Thresholding the posterior-mean death probabilities at 0.5 gives an accuracy of 0.766.

```r
p_hat <- colMeans(P_lg)
cm <- table(Predicted = as.integer(p_hat >= 0.5), Observed = yv); cm
```

```
##           Observed
## Predicted   0   1
##         0 182  50
##         1  21  46
```

```
c(accuracy = round(sum(diag(cm)) / sum(cm), 3), AUC = round(auc_fun(p_hat, yv), 3))
```

```
## accuracy      AUC
##    0.763    0.795
```

8 Inference and Prediction

Having validated the models, we now extract the inferential outputs that answer the clinical question: interval estimates for the parameters, and predictions for new patients. All quantities are read directly from the posterior draws.

8.1 Equal-tailed vs HPD Intervals

For the logistic coefficients we compare the equal-tailed 95% credible interval (2.5%–97.5% quantiles) with the 95% HPD interval; they nearly coincide because the posteriors are approximately symmetric.

```
eqt <- t(apply(sims_lg[, pars_lg], 2, quantile, c(0.025, 0.975)))
cmp <- data.frame(term = c("(Intercept)", colnames(X_red)),
                  ET_lwr = eqt[, 1], ET_upr = eqt[, 2], HPD_lwr = hpd_lg[, 1], HPD_upr = hpd_lg[, 2])
knitr::kable(cmp, digits = 3, row.names = FALSE, caption = "Equal-tailed vs HPD 95% intervals (logistic)
```

Table 9: Equal-tailed vs HPD 95% intervals (logistic)

term	ET_lwr	ET_upr	HPD_lwr	HPD_upr
(Intercept)	-1.283	-0.688	-1.268	-0.674
age	0.272	0.863	0.301	0.881
ejection_fraction	-1.069	-0.424	-1.045	-0.407
serum_creatinine	0.396	0.986	0.402	0.991
serum_sodium	-0.456	0.112	-0.445	0.122

8.2 Prediction for New Patients and a New Cohort

```
# logistic: two illustrative patient profiles (values in SD units)
newX <- rbind(high_risk = c(age = 1.5, ejection_fraction = -1.5, serum_creatinine = 2.0, serum_sodium =
              low_risk  = c(age = -1.0, ejection_fraction = 1.0, serum_creatinine = -0.5, serum_sodium =
```

```r
newX <- newX[, colnames(X_red)]
pred_prob <- apply(newX, 1, function(z) inv.logit(b0_vec + as.numeric(beta_mat %*% z)))
round(rbind(mean = colMeans(pred_prob),
            t(apply(pred_prob, 2, function(c) HPDinterval(as.mcmc(c))))), 3)
```

```
##           high_risk low_risk
## mean          0.926    0.061
## high_risk     0.858    0.977
## low_risk      0.023    0.098
```

```r
# Binomial-Beta: predicted deaths among 50 new reduced-EF patients (Beta-Binomial predictive)
pred_new <- rbinom(nrow(s_bb), 50, s_bb[, "theta[1]"])
c(mean = round(mean(pred_new), 1), HPD = round(HPDinterval(as.mcmc(pred_new)), 0))
```

```
## mean HPD1 HPD2
## 20.2 13.0 27.0
```

Interpretation. The logistic model assigns a high death probability to the high-risk profile and a low one to the low-risk profile, with full posterior uncertainty. The Binomial-Beta predictive distribution gives a full range for the number of deaths expected among 50 new reduced-EF patients, a decision-relevant output.

9 Prior Sensitivity Analysis

To confirm that the conclusions are driven by the data and not by our prior choices, we refit both models under alternative priors. The Binomial-Beta model is re-estimated under the uniform $\text{Beta}(1,1)$, the Jeffreys $\text{Beta}(0.5,0.5)$ and a mildly informative $\text{Beta}(2,2)$. The logistic model is refitted with a much tighter $\mathcal{N}(0,1)$ prior: the coefficients shrink only marginally toward zero, leaving every sign, ordering and conclusion unchanged. With roughly 300 observations the likelihood dominates the prior, so the inference is robust to the prior specification, exactly the behaviour the prior-vs-posterior plots anticipated.

```r
# Binomial-Beta under three priors
fit_bb_prior <- function(a, b) {
  cat(sprintf("model{ for(i in 1:N){ y[i] ~ dbern(theta[grp[i]]) } for(k in 1:G){ theta[k] ~ dbeta(%f,%
      file = "model_bb_p.txt")
  jags(list(y = yv, grp = grp_ef, N = N, G = 2), parameters.to.save = "theta",
       model.file = "model_bb_p.txt", n.chains = 3, n.iter = 15000, n.burnin = 3000)
}
bb_u <- fit_bb_prior(1, 1); bb_j <- fit_bb_prior(0.5, 0.5); bb_i <- fit_bb_prior(2, 2)

# logistic under a tight prior
cat("model{ for(i in 1:N){ y[i] ~ dbern(p[i]); logit(p[i]) <- b0 + inprod(X[i,],beta[]) }
      b0 ~ dnorm(0,1); for(j in 1:K){ beta[j] ~ dnorm(0,1) } }", file = "model_logit_t.txt")
```

```
M_lg_tight <- jags(dd_red, parameters.to.save = c("b0", "beta"),
                   model.file = "model_logit_t.txt", n.chains = 3, n.iter = 20000, n.burnin = 5000)
```

```
gm <- function(f) colMeans(f$BUGSoutput$sims.matrix[, c("theta[1]", "theta[2]")])
knitr::kable(round(rbind(`Beta(1,1)` = gm(bb_u), `Beta(0.5,0.5)` = gm(bb_j), `Beta(2,2)` = gm(bb_i)), 3
             caption = "Binomial-Beta: posterior means under three priors")
```

Table 10: Binomial-Beta: posterior means under three priors

	theta[1]	theta[2]
Beta(1,1)	0.401	0.202
Beta(0.5,0.5)	0.401	0.199
Beta(2,2)	0.403	0.207

```
st <- M_lg_tight$BUGSoutput$sims.matrix
knitr::kable(data.frame(term = c("(Intercept)", colnames(X_red)),
                        vague = colMeans(sims_lg[, pars_lg]), tight = colMeans(st[, pars_lg])),
             digits = 3, row.names = FALSE, caption = "Logistic: posterior means, vague vs tight prior"
```

Table 11: Logistic: posterior means, vague vs tight prior

term	vague	tight
(Intercept)	-0.973	-0.942
age	0.566	0.551
ejection_fraction	-0.735	-0.703
serum_creatinine	0.687	0.668
serum_sodium	-0.173	-0.175

Interpretation. The posterior means are virtually unchanged across priors in both models: with about 300 observations the likelihood dominates, so the conclusions are driven by the data.

10 Comparative Analysis: Frequentist vs. Bayesian

As a final check we compare the Bayesian results with their classical frequentist counterparts, to see whether the two paradigms agree on the substance for these data.

10.1 Frequentist Counterparts

The maximum-likelihood logistic regression yields coefficients almost identical to the Bayesian posterior means and the predictors it flags as significant are exactly those whose Bayesian posterior probability of a non-zero effect was near 1, while serum sodium is non-significant just as its credible interval straddled zero.

The frequentist AIC of 313.1 closely matches the Bayesian DIC of 313.4, reflecting the agreement between the two model-assessment criteria.

The convergence of the two paradigms reinforces confidence in the conclusions; the Bayesian framework adds full posterior distributions, exact probability statements and predictive distributions that the frequentist output does not provide.

```
# logistic: maximum-likelihood glm
summary(glm(f_red, data = data_std, family = binomial))
```

```
##
## Call:
## glm(formula = f_red, family = binomial, data = data_std)
##
## Coefficients:
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)        -0.9577     0.1505  -6.365 1.95e-10 ***
## age                 0.5535     0.1474   3.755 0.000174 ***
## ejection_fraction  -0.7130     0.1641  -4.344 1.40e-05 ***
## serum_creatinine    0.6754     0.1508   4.480 7.46e-06 ***
## serum_sodium       -0.1667     0.1440  -1.158 0.247063
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 375.35  on 298  degrees of freedom
## Residual deviance: 300.68  on 294  degrees of freedom
## AIC: 310.68
##
## Number of Fisher Scoring iterations: 4
```

```
# Binomial-Beta: classical test of two proportions
prop.test(table(grp_ef, yv)[, "1"], as.numeric(table(grp_ef)))
```

```
##
##  2-sample test for equality of proportions with continuity correction
##
## data:  table(grp_ef, yv)[, "1"] out of as.numeric(table(grp_ef))
## X-squared = 12.743, df = 1, p-value = 0.0003573
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  0.09622594 0.31280947
## sample estimates:
```

```
##    prop 1    prop 2
## 0.4010989 0.1965812
```

10.2 Concordance of Findings

The maximum-likelihood logistic estimates and confidence intervals closely match the Bayesian posterior means and HPD intervals, and the predictors with small p-values coincide with those whose posterior probability of a non-zero effect is near 0 or 1. The classical test of proportions confirms the Binomial-Beta result: the death proportion is significantly higher in the reduced-EF group, mirroring $P(\Delta > 0 \mid y) \approx 1$.

10.3 Methodological Comparison

In the **frequentist** view parameters are fixed constants, uncertainty is expressed through p-values and confidence intervals, prior knowledge is not used, and computation is cheap. In the **Bayesian** view parameters are random variables with full posteriors, uncertainty is expressed through credible intervals and direct probability statements, prior knowledge enters formally, and computation requires MCMC with convergence diagnostics.

10.4 Final Comparison

Both paradigms agree on the substance, reduced ejection fraction and worse renal function raise the probability of death. The Bayesian framework additionally delivers full posterior distributions, exact probability statements, predictive distributions for new patients and cohorts, and a coherent model-comparison framework via DIC.

11 Conclusions and Key Findings

This project asked what drives mortality in advanced heart failure and whether a simple model competes with a richer regression. Two models were estimated by MCMC in JAGS, a **Binomial-Beta conjugate model** stratified by ejection fraction, and a **reduced Bayesian logistic regression** on the four dominant biomarkers, and the full Bayesian pipeline was carried out: prior specification, parameter recovery on simulated data, prior-vs-posterior comparison, convergence diagnostics for both models, posterior inference with HPD intervals and odds ratios, model comparison by DIC, posterior predictive and classification checks, a prior-sensitivity analysis and a frequentist comparison.

The findings are coherent and mutually reinforcing. The Binomial-Beta model shows the death probability is substantially higher in patients with reduced ejection fraction ($P(\Delta > 0 \mid y) \approx 1$), and the analytical conjugate posterior validates the sampler exactly. The logistic model refines this picture: **ejection fraction and serum creatinine are the dominant drivers of death**, lower cardiac output and worse renal function sharply raise the odds of dying, with age and serum sodium playing smaller but credible roles. By DIC the logistic model fits better, since it exploits the continuous gradient of the biomarkers rather than a single binary split, but both models agree on the central clinical message. Posterior predictive checks reproduce

the observed mortality, the logistic model classifies patients well above chance with good calibration, the prior-sensitivity analysis confirms the results are data-driven, and classical frequentist analyses reach the same conclusions.